

Appendix A

GAMS codes

A.1 Introduction

In Chapters 5–11 different decision-making models pertaining to producers, consumers, retailers, and market/system operators are proposed and formulated. Model performances are assessed by solving and discussing examples and realistic case studies.

The objective of this Appendix is to help the reader to implement the models presented in this book. For this reason, the GAMS codes used to formulate the examples in the aforementioned chapters are included in this Appendix. We refer the interested reader to [19, 141] for further details on GAMS formulation.

Specifically, the list of examples whose GAMS codes are included in this Appendix is

1. Producer Pool Example (Section 5.7).
2. Wind Producer Example (Section 6.6).
3. Producer Futures Market Example. No Unit Unavailability (Section 7.7).
4. Producer Futures Market Example. Unit Unavailability (Section 7.8).
5. Retailer Example (Section 8.7).
6. Consumer Example (Section 9.5).
7. Market-Clearing Example (Section 10.6).
8. Market-Clearing Example with Wind Generation (Section 11.5).

A.2 GAMS code for the Producer Pool Example (Section 5.7)

```
$title POOL PRODUCER
```

```

*****
*      DATA
*****

SETS
G      Units          /g1 * g2/
T      Periods        /t1 * t3/
W      Scenarios      /w1 * w12/;

alias(W,WW);

SCALARS
beta   Weighting parameter /0/
alpha  Confidence level    /0.95/;

PARAMETERS
CP(G)   Production cost
/  g1    12
  g2    16 /

PGmax(G) Capacity
/  g1    100
  g2    50 /

PGmin(G) Minimum power ouput
/  g1    10
  g2    10 /

Rup(G)   Ramp-up ramp
/  g1    25
  g2    15 /

Rdw(G)   Ramp-down ramp
/  g1    25
  g2    15 /

PG0(G)   Initial power status
/  g1    50
  g2    25 /

PAmay(G) Maximum range power in the AGC service
/  g1    15
  g2    10 /

AD(W)    Day-ahead non-anticipativity vector
/  w1    1
  w2    1
  w3    1
  w4    0
  w5    1
  w6    1
  w7    1
  w8    0
  w9    1
  w10   1
  w11   1
  w12   0 /

AR(W)    Regulation non-anticipativity vector
/  w1    1
  w2    1
  w3    1
  w4    0
  w5    1
  w6    1
  w7    1

```

```

w8      0
w9      1
w10     1
w11     1
w12     0 /

AA(W)   Adjustment nonanticipativity vector
/ w1    1
  w2    0
  w3    1
  w4    0
  w5    1
  w6    0
  w7    1
  w8    0
  w9    1
  w10   0
  w11   1
  w12   0 /

prob(W);

TABLE lambdaD(W,T) Day-ahead price scenarios
      t1      t2      t3
w1    11.0    16.5    12.1
w2    11.0    16.5    12.1
w3    11.0    16.5    12.1
w4    11.0    16.5    12.1
w5    13.2    18.7    16.5
w6    13.2    18.7    16.5
w7    13.2    18.7    16.5
w8    13.2    18.7    16.5
w9    15.4    19.8    17.6
w10   15.4    19.8    17.6
w11   15.4    19.8    17.6
w12   15.4    19.8    17.6;

TABLE lambdaR(W,T) Regulation price scenarios
      t1      t2      t3
w1     5.5     8.5     6.5
w2     5.5     8.5     6.5
w3     5.0     7.0     5.5
w4     5.0     7.0     5.5
w5     6.5     9.5     8.0
w6     6.5     9.5     8.0
w7     6.0     8.5     7.5
w8     6.0     8.5     7.5
w9     7.0    10.5    10.5
w10    7.0    10.5    10.5
w11    6.5     9      9.5
w12    6.5     9      9.5;

TABLE lambdaA0(W,T) Adjustment price scenarios (inverse demand curve intercept)
      t1      t2      t3
w1     17     28     15
w2     10     17     13
w3     12     17     15
w4      9     14     11
w5     17     22     19
w6     14     19     15
w7     16     25     21
w8     13     17     14
w9     17     28     24
w10    14     23     20
w11    16     23     21
w12    13     19     18;

```

TABLE gammaA(W,T) Adjustment price scenarios (inverse demand curve slope)

	t1	t2	t3
w1	-0.01	-0.03	-0.02
w2	-0.01	-0.03	-0.02
w3	-0.01	-0.03	-0.02
w4	-0.01	-0.03	-0.02
w5	-0.01	-0.03	-0.02
w6	-0.01	-0.03	-0.02
w7	-0.01	-0.03	-0.02
w8	-0.01	-0.03	-0.02
w9	-0.01	-0.03	-0.02
w10	-0.01	-0.03	-0.02
w11	-0.01	-0.03	-0.02
w12	-0.01	-0.03	-0.02;

TABLE OD(W,T) Day-ahead price order

	t1	t2	t3
w1	1	1	1
w2	1	1	1
w3	1	1	1
w4	1	1	1
w5	2	2	2
w6	2	2	2
w7	2	2	2
w8	2	2	2
w9	3	3	3
w10	3	2	3
w11	3	2	3
w12	3	3	3;

prob(W)=1/card(W);

* DECLARATION OF VARIABLES

VARIABLES

z Objective function

** FIRST-STAGE VARIABLES

PD(G,T,W) Power sold in the day-ahead market

** SECOND-STAGE VARIABLES

PR(G,T,W) Power sold in the regulation market

wReg(G,T,W) 01 variable modeling if regulation service is provided

** THIRD-STAGE VARIABLES

PA(G,T,W) Power sold in the adjustment market

PG(G,T,W) Power output of the unit

** VARIABLES ASSOCIATED WITH RISK MODELING

xi Auxiliary variable used to calculate the CVaR

eta(W) Auxiliary variable used to calculate the CVaR;

POSITIVE VARIABLES PR(G,T,W), PD(G,T,W), eta(W), PA(G,T,W), EA(T,W);

BINARY VARIABLE wReg(G,T,W);

EQUATIONS

ObjectiveFunction

** DAY-AHEAD MARKET CONSTRAINTS

```

NonDecreasingDayAhead
NonAnticipativityDayAhead

** REGULATION MARKET CONSTRAINTS

MaximumReg
NonAnticipativityReg

** ADJUSTMENT MARKET CONSTRAINTS

NonAnticipativityAdjustment
PowerAdjustment

** UNIT CONSTRAINTS

Capacity
MinimumOutput
RampUp
RampDown
RampUp0
RampDown0
Power
Power0

** CVAR CONSTRAINTS

CVaR;

ObjectiveFunction.. z =e= sum(W, prob(W)*sum(T, sum(G, lambdaD(W,T)*PD(G,T,W))
+sum(G, lambdaR(W,T)*PR(G,T,W)))+(lambdaAO(W,T)*EA(T,W)+
gammaA(W,T)*EA(T,W)*EA(T,W))-sum(g, CP(G)*(PD(G,T,W)+PA(G,T,W))) )
+beta*(xi-1/(1-alpha)*sum(w, prob(W)*eta(W)));

NonDecreasingDayAhead(G,T,W,WW)$((OD(W,T)+1=OD(WW,T)) and (AD(W)=0
and (AD(WW)=0)).. PD(G,T,W)-PD(G,T,WW) =1= 0;

NonAnticipativityDayAhead(G,T,W)$ (AD(W)=1).. PD(G,T,W)-PD(G,T,W+1) =e= 0;

MaximumReg(G,T,W).. PR(G,T,W) =1= PAMax(G)*wReg(G,T,W);

NonAnticipativityReg(G,T,W)$ (AR(W)=1).. PR(G,T,W)-PR(G,T,W+1) =e= 0;

NonAnticipativityAdjustment(G,T,W)$ (AA(W)=1).. PA(G,T,W)-PA(G,T,W+1) =e= 0;

PowerAdjustment(T,W).. sum(G, PA(G,T,W)) =e= EA(T,W);

Capacity(G,T,W).. PD(G,T,W)+0.5*PR(G,T,W)+PA(G,T,W) =1= PGmax(G);

MinimumOutput(G,T,W).. PD(G,T,W)-0.5*PR(G,T,W)+PA(G,T,W) =g= PGmin(G);

RampUp(G,T,W)$ (ord(T) gt 1).. PG(G,T,W)-PG(G,T-1,W) =1=
(1-wReg(G,T,W))*Rup(G);

RampDown(G,T,W)$ (ord(T) gt 1).. PG(G,T-1,W)-PG(G,T,W) =1=
(1-wReg(G,T,W))*Rdw(G);

RampUp0(G,T,W)$ (ord(T) eq 1).. PG(G,T,W)-PG0(G) =1= (1-wReg(G,T,W))*Rup(G);

RampDown0(G,T,W)$ (ord(T) eq 1).. PG0(G)-PG(G,T,W) =1= (1-wReg(G,T,W))*Rdw(G);

Power(G,T,W)$ (ord(T) gt 1).. PD(G,T,W)+PA(G,T,W) =e=
0.5*(PG(G,T,W)+PG(G,T-1,W));

Power0(G,T,W)$ (ord(T) eq 1).. PD(G,T,W)+PA(G,T,W) =e=
0.5*(PG(G,T,W)+PG0(G));

```

```

CVar(W)..          xi-sum(T, sum(G,lambdaD(W,T)*PD(G,T,W))+
sum(G,lambdaR(W,T)*PR(G,T,W)))+(lambdaA0(W,T)*EA(T,W)+
gammaA(W,T)*EA(T,W)*EA(T,W))-sum(G,CP(G)*(PD(G,T,W)+PA(G,T,W))) ) =1= eta(W);

*****
*      MODEL
*****

MODEL PoolProducer /ALL/;
PoolProducer.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option miqcp=cplex;
SOLVE PoolProducer USING miqcp MAXIMIZING z;

```

A.3 GAMS code for the Wind Producer Example (Section 6.6)

```

$title SHORT-TERM TRADING FOR A WIND POWER PRODUCER

* DA: Day-ahead market
* AM: Adjustment market
* BM: Balancing market

*****
*      DATA
*****

SET
D  Index for DA price scenarios /d1, d2/
A  Index for price differences between DA and AM /a1, a2/
L  Index for scenarios modeling wind production in between DA and AM /l1, l2/
W  Index for scenarios modeling wind production after AM /w1, w2/
K  Index for scenarios representing imbalance price ratios /k1, k2/;

ALIAS(D,Daux);

SCALARS
dt      Time span between two consecutive time periods (h) /1/
Pmax    Wind farm capacity (MW) /100/
alfa    Confidence level /0.95/
beta    Risk-aversion parameter /0/;

TABLE P(L,W)    Wind power production
           w1    w2
l1        100    50
l2         0     40;

PARAMETERS
lambdaD(D)      Day-ahead market prices
/ d1          50
  d2          20 /

MAvsMD(A)      Price difference between day-ahead and adjustment markets
/ a1          -10
  a2           3 /

r_pos(K)      Imbalance price ratio for positive energy deviations
/ k1           1

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```

k2      0.9 /

r_neg(K)      Imbalance price ratio for negative energy deviations

/ k1      1.5
  k2      1 /

lambdaI(D,A)      Adjustment market prices

pi(D,L,A,W,K)      Scenario probabilities
/ d1.l1.a1.w1.k1      0.0098
  d1.l1.a1.w1.k2      0.0042
  d1.l1.a1.w2.k1      0.0294
  d1.l1.a1.w2.k2      0.0126
  d1.l1.a2.w1.k1      0.0182
  d1.l1.a2.w1.k2      0.0078
  d1.l1.a2.w2.k1      0.0546
  d1.l1.a2.w2.k2      0.0234
  d1.l2.a1.w1.k1      0.0343
  d1.l2.a1.w1.k2      0.0147
  d1.l2.a1.w2.k1      0.0245
  d1.l2.a1.w2.k2      0.0105
  d1.l2.a2.w1.k1      0.0637
  d1.l2.a2.w1.k2      0.0273
  d1.l2.a2.w2.k1      0.0455
  d1.l2.a2.w2.k2      0.0195
  d2.l1.a1.w1.k1      0.0147
  d2.l1.a1.w1.k2      0.0063
  d2.l1.a1.w2.k1      0.0441
  d2.l1.a1.w2.k2      0.0189
  d2.l1.a2.w1.k1      0.0273
  d2.l1.a2.w1.k2      0.0117
  d2.l1.a2.w2.k1      0.0819
  d2.l1.a2.w2.k2      0.0351
  d2.l2.a1.w1.k1      0.05145
  d2.l2.a1.w1.k2      0.02205
  d2.l2.a1.w2.k1      0.03675
  d2.l2.a1.w2.k2      0.01575
  d2.l2.a2.w1.k1      0.09555
  d2.l2.a2.w1.k2      0.04095
  d2.l2.a2.w2.k1      0.06825
  d2.l2.a2.w2.k2      0.02925 /;

lambdaI(D,A) = lambdaD(D) + MAVsMD(A);

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES

z                      Objective function value
profit(D,L,A,W,K)      Profit per scenario

** FIRST-STAGE VARIABLES

Pd(D)                  Power sold in the day-ahead market

** SECOND-STAGE VARIABLES

Pa(D,L)                Power traded in the adjustment market
Ps(D,L)                Final power schedule

** THIRD-STAGE VARIABLES

desvP(D,L,W)           Positive energy deviation
desvN(D,L,W)           Negative energy deviation

```

```

** VARIABLES ASSOCIATED WITH RISK MODELING

CVaR                Conditional value-at-risk
var                 Value-at-risk
eta(D,L,A,W,K)      Auxiliary variable

*****
*      MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES Pd(D), Ps(D,L), desvP(D,L,W), desvN(D,L,W), eta(D,L,A,W,K);

Pd.up(D) = Pmax;
Ps.up(D,L) = Pmax;

*****
*      EQUATIONS
*****

EQUATIONS

OF                Objective function
CVaRfunction      CVaR definition
PROFITfunction(D,L,A,W,K) Profit definition
DesvDEF(D,L,W)    Definition of energy imbalances
MAXdesvPOS(D,L,W) Maximum positive energy deviation
MAXdesvNEG(D,L,W) Maximum negative energy deviation
PsDEF(D,L)        Computation of final power schedule
curve(D,Daux)     Non-decreasing condition of offering curves
noANTICIPd(D,Daux) Non-anticipativity constraints for the DA market
CVaRrest(D,L,A,W,K) Constraints associated with risk modeling;

OF.. z =e= sum((D,L,A,W,K), pi(D,L,A,W,K)*profit(D,L,A,W,K)) + beta*CVaR;

CVaRfunction..

CVaR =e= var - (1/(1-alfa))*sum((D,L,A,W,K), pi(D,L,A,W,K)*eta(D,L,A,W,K));

PROFITfunction(D,L,A,W,K)..

profit(D,L,A,W,K) =e= dt*lambdaD(D)*Pd(D)
                    + dt*lambdaI(D,A)*Pa(D,L)
                    + lambdaD(D)*r_pos(K)*desvP(D,L,W)
                    - lambdaD(D)*r_neg(K)*desvN(D,L,W);

DesvDEF(D,L,W).. desvP(D,L,W)-desvN(D,L,W) =e= dt*(P(L,W) - Ps(D,L));

MAXdesvPOS(D,L,W).. desvP(D,L,W) =l= P(L,W)*dt;

MAXdesvNEG(D,L,W).. desvN(D,L,W) =l= Pmax*dt;

PsDEF(D,L).. Ps(D,L) =e= Pd(D) + Pa(D,L);

curve(D,Daux)$(lambdaD(Daux) GT lambdaD(D)).. Pd(D)-Pd(Daux) =l= 0;

noANTICIPd(D,Daux)$(lambdaD(Daux) EQ lambdaD(D)).. Pd(D) =e= Pd(Daux);

CVaRrest(D,L,A,W,K).. var - profit(D,L,A,W,K) - eta(D,L,A,W,K) =l= 0;

*****
*      MODEL
*****

MODEL WindProdProblem /ALL/;
OPTION iterlim = 1e8;
OPTION reslim = 1e10;

```

```

Option lp=cplex;
WindProdProblem.optcr=0;
SOLVE WindProdProblem USING lp MAXIMIZING z;

```

A.4 GAMS code for the Producer Futures Market Example. No Unit Unavailability (Section 7.7)

```

$title PRODUCER (NO UNIT UNAVAILABILITY)

*****
* DATA
*****

SETS
G      Generators                /g1 * g2/
W      Scenarios                 /w1 * w4/
T      Periods                  /t1 * t4/
F      Forward contracts         /f1 * f2/
J      Blocks in the forward contracting curve /j1 * j2/
SPeak(T) peak hours;

SPeak(T)=no; SPeak('t2')=yes; SPeak('t4')=yes;

SCALARS
beta    Weighting factor /0/
alpha   Confidence level /0.95/;

PARAMETERS
CG(G)    Production cost
/ g1      31
  g2      32 /

Pmax(G)   Capacity of the units
/ g1      300
  g2      250 /

Pmin(G)   Minimum output of the units
/ g1      0
  g2      0 /

H(G)      Relationship between the production in peak and off-peak hours
/ g1      0.33
  g2      0.33 /

d(T)      Duration in hours of every period t
EPC(T)    Energy previously contracted
prob(W)   Scenarios probability;

d(T)=1;
EPC(T)=0;
prob(W)=0.25;

TABLE PFmax(F,J) Upper limit of the forward contracting blocks
      j1      j2
f1     100     100
f2     100     100;

TABLE lambdaF(F,J) Price of the forward contracting blocks

```

```

      j1      j2
f1      35      34
f2      34.5    32;

TABLE lambdaP(W,T) Pool price
      t1      t2      t3      t4
w1      33      43      29      36
w2      32      39      26      33
w3      34      48      28      37
w4      33      45      26      39;

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES
      z                      Objective function

** FIRST-STAGE VARIABLES

      PF(F,J)              Power sold through forward contracts

** SECOND-STAGE VARIABLES

      EP(T,W)              Energy traded in the pool
      EG(G,T,W)           Energy generated by the units

** VARIABLES ASSOCIATED WITH RISK MODELING

      xi                  Auxiliary variable used to calculate the CVaR
      eta(W)             Auxiliary variable used to calculate the CVaR

*****
*      MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES    PF(F,J),EP(T,W),EG(G,T,W),eta(W);

*****
*      EQUATIONS
*****

EQUATIONS
ObjetiveFunction

** FORWARD CONTRACT CONSTRAINTS

LimitContractMax

** UNIT CONSTRAINTS

LimitGenMin
LimitGenMax
RelPeakBase

** ENERGY BALANCE

EnergyBalance

** CVAR CONSTRAINTS

CVaRConstraint;

ObjetiveFunction..      z =e= sum(W,prob(W)*sum(T,sum(F,sum(J,PF(F,J)*
lambdaP(F,J))))*d(T)+lambdaP(W,T)*EP(T,W)-sum(G,CG(G)*EG(G,T,W)))+
beta*(xi-1/(1-alpha)*sum(W,prob(W)*eta(W)));

```

```

LimitContractMax(F,J)..      PF(F,J) =l= PFmax(F,J);

LimitGenMin(G,T,W)..        EG(G,T,W) =l= Pmax(G)*d(T);

LimitGenMax(G,T,W)..        EG(G,T,W) =g= Pmin(G)*d(T);

RelPeakBase(G,T,W)$SPeak(T).. EG(G,T-1,W) =g= H(G)*EG(G,T,W);

EnergyBalance(T,W)..        sum(G,EG(G,T,W)) =e= EP(T,W)+
sum(F,sum(J,PF(F,J)*d(T)))+EPC(T);

CVarConstraint(W)..          xi-(sum(T,sum(F,sum(J,PF(F,J)*lambdaF(F,J)))*d(T)
+lambdaP(W,T)*EP(T,W)-sum(G,CG(G)*EG(G,T,W)))) =l= eta(W);

*****
*                               MODEL
*****

MODEL Producer /ALL/;
Producer.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option lp=cplex;
SOLVE Producer USING lp MAXIMIZING z;

```

A.5 GAMS code for the Producer Futures Market Example. Unit Unavailability (Section 7.8)

```

$title PRODUCER UNIT AVAILABILITY

*****
*                               DATA
*****

SETS
G      Generators                /g1 * g2/
W      Scenarios                 /w1 * w20/
T      Periods                  /t1 * t4/
F      Forward contracts        /f1 * f2/
J      Blocks in the forward contracting curve /j1 * j2/
SPeak(T) peak hours;

SPeak(T)=no; SPeak('t2')=yes; SPeak('t4')=yes;

SCALARS
beta    Weighting factor /0/
alpha   Confidence level /0.95/;

PARAMETERS
CG(G)    Production cost
/ g1     31
  g2     32 /

Pmax(G)  Capacity of the units
/ g1     300
  g2     250 /

Pmin(G)  Minimum output of the units
/ g1     0

```

```

g2      0 /

H(G)    Relationship between the production in peak and off-peak hours
/ g1     0.33
  g2     0.33 /

prob(W) Scenarios probability
/ w1     0.24
  w2     0.24
  w3     0.24
  w4     0.24
  w5     0.0025
  w6     0.0025
  w7     0.0025
  w8     0.0025
  w9     0.0025
  w10    0.0025
  w11    0.0025
  w12    0.0025
  w13    0.0025
  w14    0.0025
  w15    0.0025
  w16    0.0025
  w17    0.0025
  w18    0.0025
  w19    0.0025
  w20    0.0025/

d(T)    Duration in hours of every period t
EPC(T)  Energy previously contracted;

d(T)=1;
EPC(T)=0;

TABLE PFmax(F,J) Upper limit of the forward contracting blocks
      j1      j2
f1    100     100
f2    100     100;

TABLE lambdaF(F,J) Price of the forward contracting blocks
      j1      j2
f1     35      34
f2    34.5     32;

TABLE lambdaP(W,T) pool price
      t1      t2      t3      t4
w1    33      43      29      36
w2    32      39      26      33
w3    34      48      28      37
w4    33      45      26      39
w5    33      43      29      36
w6    32      39      26      33
w7    34      48      28      37
w8    33      45      26      39
w9    33      43      29      36
w10   32      39      26      33
w11   34      48      28      37
w12   33      45      26      39
w13   33      43      29      36
w14   32      39      26      33
w15   34      48      28      37
w16   33      45      26      39
w17   33      43      29      36
w18   32      39      26      33
w19   34      48      28      37
w20   33      45      26      39
;

```

```

TABLE a_aux(w,T) availability of unit 2
      t1      t2      t3      t4
w1      1      1      1      1
w2      1      1      1      1
w3      1      1      1      1
w4      1      1      1      1
w5      0      1      1      1
w6      0      1      1      1
w7      0      1      1      1
w8      0      1      1      1
w9      1      0      1      1
w10     1      0      1      1
w11     1      0      1      1
w12     1      0      1      1
w13     1      1      0      1
w14     1      1      0      1
w15     1      1      0      1
w16     1      1      0      1
w17     1      1      1      0
w18     1      1      1      0
w19     1      1      1      0
w20     1      1      1      0
;
PARAMETER a(G,W,T);
a('g1',w,t)=1;
a('g2',w,t)=a_aux(w,t);

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES
      z                      Objective function

** FIRST-STAGE VARIABLES

      PF(F,J)              Power sold through forward contracts

** SECOND-STAGE VARIABLES

      EP(T,W)              Energy traded in the pool
      EG(G,T,W)            Energy generated by the units

** VARIABLES ASSOCIATED WITH RISK MODELING

      xi                   Auxiliary variable used to calculate the CVaR
      eta(W)               Auxiliary variable used to calculate the CVaR

*****
*      MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES   PF(F,J),EP(T,W),EG(G,T,W),eta(W);

*****
*      EQUATIONS
*****

EQUATIONS

ObjectiveFunction

** FORWARD CONTRACT CONSTRAINTS

LimitContractMax

** UNIT CONSTRAINTS

```

```

LimitGenMin
LimitGenMax
RelPeakBase

** ARBITRAGE CONSTRAINTS

LimitBuyPool

** ENERGY BALANCE

EnergyBalance

** CVAR CONSTRAINTS

CVarConstraint;

ObjectiveFunction..          z=e=sum(W,prob(W)*sum(T,lambdaP(W,T)*EP(T,W)+
sum(F,sum(J,PF(F,J)*lambdaF(F,J))) *d(T)-sum(G,CG(G)*EG(G,T,W))))+
beta*(xi-1/(1-alpha)*sum(W,prob(W)*eta(W)));

LimitContractMax(F,J)..      PF(F,J)=1=PFmax(F,J);

LimitGenMin(G,T,W)..          EG(G,T,W)=1= PMAX(G)*d(T)*a(G,W,T);

LimitGenMax(G,T,W)..          EG(G,T,W)=g= PMIN(G)*d(T)*a(G,W,T);

LimitBuyPool(T,W)..           EP(T,W)=g=(a('g2',W,T)-1)*
sum(F,sum(J,PF(F,J)*d(T)));

RelPeakBase(G,T,W)$SPeak(T).. EG(G,T-1,W)=g=H(G)*EG(G,T,W);

EnergyBalance(T,W)..          sum(g,EG(G,T,W))=e=EP(T,W)+
sum(F,sum(J,PF(F,J)*d(T)));

CVarConstraint(W)..            -(sum(T,lambdaP(W,T)*EP(T,W)+
sum(F,sum(J,PF(F,J)*lambdaF(F,J))) *d(T)-sum(G,CG(G)*EG(G,T,W))))+xi-eta(W)=1=0;

*****
*           MODEL
*****

MODEL ProducerAvailability /ALL/;
ProducerAvailability.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option lp=cplex;
SOLVE ProducerAvailability USING lp MAXIMIZING z;

```

A.6 GAMS code for the Retailer Example (Section 8.7)

```

$title RETAILER

*****
*           DATA
*****

SETS

```

```

W      Scenarios                               /w1 * w4/
T      Periods                               /t1 * t2/
F      Forward contracts                     /f1 * f2/
E      Client groups                       /e1 * e1/
J      Blocks in the forward contracting curve /j1 * j2/
I      Blocks in the piece-quota curve       /i1 * i3/
Ft(F,T) Forward contracts available in period t;

Ft(F,T)=yes;

SCALARS

lambdaRmax      Maximum selling price /120/
lambdaRmin      Minimum selling price /20/
beta            Weighting factor      /0/
alpha           Confidence level      /0.95/;

PARAMETERS

EPC(T)          Energy previously contracted
ED(T,E,W)       Stochastic demand of the clients
ER(T,E,I,W)     Stochastic demand supplied by the retailer
d(T)            Duration in hours of every period t
prob(W)         Scenario probability;

TABLE lambdaF(F,J) Price of the forward contracting blocks
      j1      j2
f1      66.00  69.30
f2      67.00  70.35;

TABLE PFmax(F,J) Upper limit of the forward contracting blocks
      j1      j2
f1      20     20
f2      20     20;

TABLE k(I,E) Demand supplied in each price-quota curve block (per unit)
      e1
i1      1.00
i2      0.35
i3      0.15;

TABLE lambdaRI(I,E) Price of each price-quota curve block
      e1
i1      65.0
i2      75.0
i3      85.0;

TABLE lambdaP(W,T) Pool prices
      t1      t2
w1      60     52
w2      65     55
w3      74     68
w4      70     61;

ED('t1',E,'w1')=350; ED('t2',E,'w1')=325; ED('t1',E,'w2')=365;
ED('t2',E,'w2')=335; ED('t1',E,'w3')=375; ED('t2',E,'w3')=345;
ED('t1',E,'w4')=360; ED('t2',E,'w4')=340;
d(T)=1;
prob(W)=1/3;
ER(T,E,I,W)=ED(T,E,W)*k(I,E);
EPC(T)=0;

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES

```

```

z                Objective function

** FIRST-STAGE VARIABLES

PF(F,J)          Power bought from forward contracts
lambdaR(E,I)     Auxiliary variable for computing the selling price
v(E,I)           Selection of block I in price-quota curve of group E

** SECOND-STAGE VARIABLES

EP(T,W)          Energy traded in the pool

** VARIABLES ASSOCIATED WITH RISK MODELING

xi               Auxiliary variable used to calculate the CVaR
eta(W)           Auxiliary variable used to compute the CVaR;

*****
*               MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES PF(F,J),eta(W);
BINARY VARIABLES v(E,I);

*****
*               EQUATIONS
*****

EQUATIONS

ObjectiveFunction

** FORWARD CONTRACT CONSTRAINTS

LimitContractMax

** CLIENT CONSTRAINTS

LimitPriceMin
LimitPriceMax
PriceQuota

** ENERGY BALANCE CONSTRAINTS

EnergyBalance

** CVAR CONSTRAINTS

CVaRConstraint;

ObjectiveFunction..      z =e= sum(W,prob(w))*sum(T,sum((E,I),ER(T,E,I,W)*
lambdaR(E,I))-lambdaP(W,T)*EP(T,W)-sum(F$Ft(F,T),sum(J,PF(F,J)*
lambdaF(F,J))*d(T))+beta*(xi-1/(1-alpha)*sum(W,prob(W)*eta(W)));

LimitContractMax(F,J).. PF(F,J) =l= PFmax(F,J);

LimitPriceMin(E,I)$(ord(I) gt 1)..      lambdaR(E,I) =g=
lambdaRI(I-1,E)*v(E,I);

LimitPriceMax(E,I).. lambdaR(E,I) =l= lambdaRI(I,E)*v(E,I);

PriceQuota(E)..          sum(I,v(E,I)) =e= 1;

EnergyBalance(T,W)..      sum((E,I),ER(T,E,I,W)*v(E,I)) =e= EP(T,W)+
sum(F$Ft(F,T),sum(J,PF(F,J)*d(T)))+EPC(T);

```

```

CVarConstraint(W).. xi-(sum(T,sum((E,I),ER(T,E,I,W)*lambdaR(E,I))
-lambdaP(W,T)*EP(T,W)-sum(F$Ft(F,T),sum(J,PF(F,J)*lambdaF(F,J))*d(T)))
=l= eta(W);

*****
*           MODEL
*****

MODEL Retailer /ALL/;
Retailer.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option mip=cplex;
SOLVE Retailer USING mip MAXIMIZING z;

```

A.7 GAMS code for the Consumer Example (Section 9.5)

```

$title CONSUMER

*****
*           DATA
*****

SETS
B      Bilateral contracts           /b1 * b6/
K      Stages                       /k1 * k4/
W      Scenarios                     /w1 * w16/
T      Periods                      /t1 * t4/
N      Blocks in the piecewise linear production cost /n1/
Bt(B,T) Mapping of bilateral contracts active in period T
Kb(B,K) Mapping of decisions on bilateral contracts made in stage K;

Bt(B,T)=no; Bt('b1',t)=yes; Bt('b2',t)=yes; Bt('b3','t1')=yes;
Bt('b4','t2')=yes; Bt('b5','t3')=yes; Bt('b6','t4')=yes;

Kb(B,K)=no; Kb('b1','k1')=yes; Kb('b2','k1')=yes; Kb('b3','k1')=yes;
Kb('b4','k2')=yes; Kb('b5','k3')=yes; Kb('b6','k4')=yes;

SCALARS

beta   Weighting factor /0/
alpha  Confidence level /0.95/;

PARAMETERS

ED(T) Demand
/ t1      200
  t2      250
  t3      225
  t4      275 /

lambdaBo(B) Reference price of electricity for bilateral contracts
/ b1      48.5
  b2      50.0
  b3      49.5
  b4      51.0
  b5      49.0
  b6      50.0 /

```

```

PBmax(B) Upper bound of bilateral contract purchases
/ b1      75
  b2      75
  b3      50
  b4      50
  b5      50
  b6      50 /

PBmin(B) Lower bound of bilateral contract purchases
/ b1      20
  b2      20
  b3      20
  b4      20
  b5      20
  b6      20 /

EGT(N) Blocks of the lineal approximation of the variable costs
/ n1      100 /

SG(N) Slope of the approximation of the variable costs
/ n1      45 /

lambdaB(B,T,W) Price of electricity for bilateral contracts
d(T)           Duration in hours of every period t
EPC(T)         Energy previously contracted
prob(W)        Scenario probability;

TABLE lambdaP(W,T) Pool price
      t1      t2      t3      t4
w1     44     47     45     45
w2     44     47     45     47
w3     44     47     46     47
w4     44     47     46     49
w5     44     44     42     42
w6     44     44     42     45
w7     44     44     44     45
w8     44     44     44     46
w9     50     52     50     51
w10    50     52     50     55
w11    50     52     51     52
w12    50     52     51     53
w13    50     49     47     49
w14    50     49     47     50
w15    50     49     48     50
w16    50     49     48     53;

TABLE A(W,K) Non-anticipativity matrix
      k1      k2      k3      k4
w1      1      1      1      1
w2      1      1      1      0
w3      1      1      1      1
w4      1      1      0      0
w5      1      1      1      1
w6      1      1      1      0
w7      1      1      1      1
w8      1      0      0      0
w9      1      1      1      1
w10     1      1      1      0
w11     1      1      1      1
w12     1      1      0      0
w13     1      1      1      1
w14     1      1      1      0
w15     1      1      1      1;

d(T)=1;
EPC(T)=0;

```

```

lambdaB(B,T,W)=(lambdaBo(B)+lambdaP(W,T))/2;
prob(W)=0.0625;

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES

    z                Objective function value
    PB(B,W)          Power purchased from bilateral contracts
    s(B,W)           Selection of bilateral contracts
    EG(N,T,W)        Energy produced by the self-production unit
    EP(T,W)          Energy traded in the pool

** VARIABLES ASSOCIATED WITH RISK MODELING

    xi              Auxiliary variable used to calculate the CVaR
    eta(W)          Auxiliary variable used to calculate the CVaR;

*****
*      MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES EG(N,T,W),eta(W);
BINARY VARIABLES s(B,W);

*****
*      EQUATIONS
*****

EQUATIONS

ObjectiveFunction

** BILATERAL CONTRACT CONSTRAINTS

LimitContractMax
LimitContractMin

** SELF-PRODUCTION CONSTRAINTS

SelfProduction1
SelfProduction

** ENERGY BALANCE

EnergyBalance

** ARBITRAGE CONSTRAINTS

Arbitrage

** NON-ANTICIPATIVITY CONSTRAINTS

Non_Anticipativity

** CVAR CONSTRAINTS

CVaR;

ObjectiveFunction..          z =e= sum(W,prob(W)*sum(T,sum(B$Bt(B,T),
lambdaB(B,T,W)*PB(B,W)*d(T))+lambdaP(W,T)*EP(T,W)+sum(N,SG(N)*EG(N,T,W))))+
beta*(xi+1/(1-alpha)*sum(W,prob(W)*eta(W)));

LimitContractMax(B,W)..     PB(B,W) =l= PBmax(B)*s(B,W);

```

```

LimitContractMin(B,W)..          PB(B,W) =g= PBmin(B)*s(B,W);

SelfProduction1(N,T,W)$ (ord(N) eq 1)..          EG(N,T,W) =l= EGT(N);

SelfProduction(N,T,W)$ (ord(N) gt 1)..          EG(N,T,W) =l= EGT(N)-EGT(N-1);

EnergyBalance(T,W)..          sum(N,EG(N,T,W))+EP(T,W)+sum(B,$Bt(B,T),
PB(B,W)*d(T)) =e= ED(T)-EPC(T);

Arbitrage(T,W)..          sum(N,EG(N,T,W)) =g= -EP(T,W);

Non_Anticipativity(B,W,K)$ (KB(B,K) and (A(W,K)=1) and (ord(W) lt card(W)))..
PB(B,W) =e= PB(B,W+1);

CVar(W)..          sum(T,sum(B,$Bt(B,T),lambdaB(B,T,W)*PB(B,W)*d(T))
+lambdaP(W,T)*EP(T,W)+sum(N,SG(N)*EG(N,T,W)))-xi-eta(W) =l= 0;

*****
*      MODEL
*****

MODEL Consumer /ALL/;
Consumer.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option mip=cplex;
SOLVE Consumer USING mip MINIMIZING z;

```

A.8 GAMS code for the Market-Clearing Example (Section 10.6)

```

$title MARKET CLEARING UNDER EQUIPMENT FAILURES

*****
*      DATA
*****

SETS
N   Nodes          /n1 * n3/
I   Units          /i1 * i3/
J   Loads          /j1 * j1/
T   Periods        /t1 * t4/
W   Scenarios      /w0 * w8/
M   Offer blocks   /m1 * m1/
S   System states  /s0*s2/;

ALIAS(N,R);

SCALARS
NT      Number of time periods /4/,
dt      Duration of time periods /1/
x       Reactance of transmission lines /0.13/
Pbase   Power base /41/;

TABLE GDATA(I,*) Generation technical data
      PMIN  PMAX  Istatus  Pini
i1     10    100    0        0
i2     10    100    0        0
i3     10    50    0        0;

```

TABLE RDATA_G(I,T,*) Generation-side reserve data

	Rspin_max_up	Rspin_max_dn	Rnspin_max
i1.t1*t4	90	90	100
i2.t1*t4	90	90	100
i3.t1*t4	40	40	50;

TABLE CostBlock(I,T,M) Offer cost of energy blocks

	m1
i1.t1*t4	30
i2.t1*t4	40
i3.t1*t4	20;

TABLE WidthBlock(I,T,M) Width of energy blocks

	m1
i1.t1*t4	100
i2.t1*t4	100
i3.t1*t4	50;

TABLE Cr_schG(I,T,*) Generation-side reserve offer cost

	up	dn	NS
i1.t1*t4	5	5	4.5
i2.t1*t4	7	7	5.5
i3.t1*t4	8	8	7;

TABLE ML(J,N) Mapping of the set of loads into the set of nodes

	n1	n2	n3
j1	0	0	1;

TABLE NetState(N,N,S) Network states

	s0	s1	s2
n1.n1	0	0	0
n1.n2	1	1	1
n1.n3	1	1	0
n2.n1	1	1	1
n2.n2	0	0	0
n2.n3	1	1	1
n3.n1	1	1	0
n3.n2	1	1	1
n3.n3	0	0	0;

TABLE GenState(I,N,S) States of unit availability

	s0	s1	s2
i1.n1	1	1	1
i1.n2	0	0	0
i1.n3	0	0	0
i2.n1	0	0	0
i2.n2	1	1	1
i2.n3	0	0	0
i3.n1	0	0	0
i3.n2	0	0	0
i3.n3	1	0	1;

PARAMETERS

elast_limits(J,T,*) Elasticity limits of loads

/	j1.t1.sup	30
	j1.t2.sup	80
	j1.t3.sup	110
	j1.t4.sup	40
	j1.t1.inf	30
	j1.t2.inf	80
	j1.t3.inf	110
	j1.t4.inf	40 /

RDATA_D(J,T,*) Demand-side reserve data

```

/  j1.t1.Rspin_max_up  3
   j1.t2.Rspin_max_up  8
   j1.t3.Rspin_max_up  11
   j1.t4.Rspin_max_up  4
   j1.t1.Rspin_max_dn  3
   j1.t2.Rspin_max_dn  8
   j1.t3.Rspin_max_dn  11
   j1.t4.Rspin_max_dn  4 /

prob(W) Scenario w probability
/  w0  0.98807312
   w1  0.00199015
   w2  0.00199015
   w3  0.00199015
   w4  0.00199015
   w5  0.00099157
   w6  0.00099157
   w7  0.00099157
   w8  0.00099157 /

TAU(W) Period in which the contingency defining scenario w occurs
/  w0  5
   w1  1
   w2  2
   w3  3
   w4  4
   w5  1
   w6  2
   w7  3
   w8  4 /

NumO(I,T)      Number of energy blocks offered by each unit
lambdaSU(I,T)  Start-up offer cost
lambdaL(J,T)   Demand utility
Cr_schD(J,T,*) Demand-side reserve offer cost
Vlol(J,T)      Value of lost load
b(N,N)         Imaginary parts of the admittance of lines
fmax(N,N)      Transmission capacity limits
conec(N,N,T,W) Network state in time period t and scenario w
MG(I,N,T,W)    Mapping of the set of units into the set of nodes
G(I,T,W)       Set of available units per scenario and period;

NumO(I,T) = 1;
lambdaSU(I,T) = 100;
b(N,R)$(ord(N) ne ord(R)) = -1/x;
G(I,T,W) = 1;
lambdaL(J,T) = 0;
Cr_schD(J,T,'up') = 70;
Cr_schD(J,T,'dn') = 70;
Vlol(J,T) = 1000;
fmax(N,R)$(ord(N) ne ord(R)) = 55;

loop(T,
  loop(W,
    if(ord(T) < TAU(W),
      conec(N,R,T,W) = NetState(N,R,'s0');
      MG(I,N,T,W) = GenState(I,N,'s0');
    else
      if(ord(W) <= 5,
        conec(N,R,T,W) = NetState(N,R,'s1');
        MG(I,N,T,W) = GenState(I,N,'s1');
        G('i3',T,W) = 0;
      else
        conec(N,R,T,W) = NetState(N,R,'s2');
        MG(I,N,T,W) = GenState(I,N,'s2');
      );
    );
  );
);

```

```

    );
);

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES
    EC                                Expected cost (objective function value)

** FIRST-STAGE VARIABLES
    Ps(I,T)                          Power output schedule
    Ls(J,T)                          Demand schedule
    R_spin_schG_d(I,T)               Generation-side spinning reserve schedule (downward)
    R_spin_schG_u(I,T)               Generation-side spinning reserve schedule (upward)
    R_spin_schD_d(J,T)               Demand-side spinning reserve schedule (downward)
    R_spin_schD_u(J,T)               Demand-side spinning reserve schedule (upward)
    R_nspin_schG(I,T)                Generation-side non-spinning reserve schedule
    u(I,T)                           Scheduled commitment status

** SECOND-STAGE VARIABLES
    Lc(J,T,W)                        Power consumption in real time
    Lshed(J,T,W)                     Amount of involuntarily load shed
    Pg(I,T,W)                        Generator power output in real time
    pgblock(I,T,W,M)                 Power produced from energy blocks
    r_spin_depG_d(I,T,W)              Generation-side spinning reserve deployment (downward)
    r_spin_depG_u(I,T,W)              Generation-side spinning reserve deployment (upward)
    r_spin_depD_d(J,T,W)              Demand-side spinning reserve deployment (downward)
    r_spin_depD_u(J,T,W)              Demand-side spinning reserve deployment (upward)
    r_nspin_depG(I,T,W)               Generation-side non-spinning reserve deployment
    angle(N,T,W)                     Voltage angle
    Pinj(N,T,W)                      Power injection
    f(N,R,T,W)                       Power flow
    v(I,T,W)                          Real-time commitment status
    Csu(I,T,W)                       Start-up cost;

*****
*      MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES
    Csu(I,T,W), Lshed(J,T,W), r_spin_depG_d(I,T,W), r_spin_depG_u(I,T,W),
    r_nspin_depG(I,T,W), r_spin_depD_d(J,T,W), r_spin_depD_u(J,T,W),
    R_spin_schG_d(I,T), R_spin_schG_u(I,T), R_spin_schD_d(J,T), R_spin_schD_u(J,T),
    R_nspin_schG(I,T), pgblock(I,T,W,M);

BINARY VARIABLES u(I,T), v(I,T,W);

** Elasticity limits of demand
    Ls.up(J,T)= elast_limits(J,T,'sup');
    Ls.lo(J,T)= elast_limits(J,T,'inf');

** Reference node
    angle.fx('N1',T,W)=0;

** Non-anticipativity constraints
    Lshed.fx(J,T,W)$ (ord(T) LT TAU(W))= 0;
    r_spin_depG_d.fx(I,T,W)$ (ord(T) LT TAU(W))= 0;
    r_spin_depG_u.fx(I,T,W)$ (ord(T) LT TAU(W))= 0;
    r_spin_depD_d.fx(J,T,W)$ (ord(T) LT TAU(W))= 0;
    r_spin_depD_u.fx(J,T,W)$ (ord(T) LT TAU(W))= 0;
    r_nspin_depG.fx(I,T,W)$ (ord(T) LT TAU(W))= 0;

*****
*      EQUATIONS
*****

```

```

EQUATIONS

ECFunction      Objective function

** ELECTRICITY MARKET CONSTRAINTS

** Production limits
MAX_PROD(I,T)
MIN_PROD(I,T)

** Market equilibria
MARKET_EQU(T)

** Scheduled reserve determination constraints

** Generation side
RESERVE_SCH_SPIN_G_u(I,T) Spinning up
RESERVE_SCH_SPIN_G_d(I,T) Spinning down
RESERVE_SCH_NSPIN_G(I,T) Non-spinning

** Demand side
RESERVE_SCH_SPIN_D_u(J,T) Spinning up
RESERVE_SCH_SPIN_D_d(J,T) Spinning down

** REAL-TIME OPERATING CONSTRAINTS

** Start-up costs
SUo(I,W)      T = 1
SU(I,T,W)     T > 1

** Generation limits
GL1(I,T,W)
GL2(I,T,W)

** Decomposition of generator power outputs into blocks
GL3(I,T,W,M)
GL4(I,T,W)

** Involuntary load shedding constraints
LIMIT_ENS(J,T,W)

** Network constraints
PB(N,T,W)      Power balance
DEF_Pinj(N,T,W) Definition of power injection
DEF_FLOW(N,R,T,W) Defintion of power flows
MAX_CAP(N,R,T,W) Transmission capacity

** LINKING CONSTRAINTS
Pgenerated(I,T,W) Composition of generator power outputs
Pconsumed(J,T,W) Composition of the power consumption

** Deployed reserve determination constraints

** Generation side
RESERVE_DPL_SPIN_G_u(I,T,W) Spinning up
RESERVE_DPL_SPIN_G_d(I,T,W) Spinning down
RESERVE_DPL_NSPIN_G(I,T,W) Non-spinning

** Demand side
RESERVE_DPL_SPIN_D_u(J,T,W) Spinning up
RESERVE_DPL_SPIN_D_d(J,T,W) Spinning down;

ECFunction..

EC =e= dt*sum((I,T),

```

```

Cr_schG(I,T,'up')* R_spin_schG_u(I,T)+ Cr_schG(I,T,'dn')* R_spin_schG_d(I,T)
+ Cr_schG(I,T,'NS')* R_nspin_schG(I,T)) + dt*sum((J,T),
Cr_schD(J,T,'up')* R_spin_schD_u(J,T)+ Cr_schD(J,T,'dn')* R_spin_schD_d(J,T))
+ dt*sum((T,W),
prob(W)*(sum((I,M)$ (ord(M) LE Num0(I,T)), CostBlock(I,T,M)* pgblock(I,T,W,M))
- sum(J, lambdaL(J,T)* Lc(J,T,W))
+ sum(J, V1ol(J,T)*Lshed(J,T,W))))
+ sum((I,T,W),prob(W)*Csu(I,T,W)));

MIN_PROD(I,T).. Ps(I,T) =g= GDATA(I,'PMIN')* u(I,T);

MAX_PROD(I,T).. Ps(I,T) =l= GDATA(I,'PMAX')*u(I,T);

MARKET_EQU(T).. sum(I, Ps(I,T)) =e= sum(J, Ls(J,T));

RESERVE_SCH_SPIN_G_u(I,T)..

R_spin_schG_u(I,T) =l= RDATA_G(I,T,'Rspin_max_up')*u(I,T);

RESERVE_SCH_SPIN_G_d(I,T)..

R_spin_schG_d(I,T) =l= RDATA_G(I,T,'Rspin_max_dn')*u(I,T);

RESERVE_SCH_NSPIN_G(I,T)..

R_nspin_schG(I,T) =l= RDATA_G(I,T,'Rnspin_max')*(1-u(I,T));

RESERVE_SCH_SPIN_D_u(J,T).. R_spin_schD_u(J,T) =l= RDATA_D(J,T,'Rspin_max_up');
RESERVE_SCH_SPIN_D_d(J,T).. R_spin_schD_d(J,T) =l= RDATA_D(J,T,'Rspin_max_dn');
SUo(I,W).. Csu(I,'t1',W) =g= lambdaSU(I,'t1')*(v(I,'t1',W)-GDATA(I,'Istatus'));
SU(I,T,W)$ (ord(T) GT 1).. Csu(I,T,W) =g= lambdaSU(I,T)*(v(I,T,W)-v(I,T-1,W));
GL1(I,T,W).. Pg(I,T,W) =g= GDATA(I,'PMIN')* v(I,T,W);
GL2(I,T,W).. Pg(I,T,W) =l= GDATA(I,'PMAX')*v(I,T,W);
GL3(I,T,W,M)$ (ord(M) LE Num0(I,T)).. pgblock(I,T,W,M) =l= WidthBlock(I,T,M);
GL4(I,T,W).. Pg(I,T,W) =e= sum(M$(ord(M) LE Num0(I,T)), pgblock(I,T,W,M));
LIMIT_ENS(J,T,W).. Lshed(J,T,W)=l= Lc(J,T,W);
PB(N,T,W).. sum(I$(MG(I,N,T,W) EQ 1),Pg(I,T,W))- sum(J$(ML(J,N) EQ 1),
Lc(J,T,W)-Lshed(J,T,W))-Pinj(N,T,W) =e= 0;
DEF_Pinj(N,T,W).. Pinj(N,T,W) =e= sum(R$(conec(N,R,T,W) EQ 1),f(N,R,T,W));
DEF_FLOW(N,R,T,W)$ (conec(N,R,T,W) EQ 1)..

f(N,R,T,W)=e= -b(N,R)*(angle(N,T,W)-angle(R,T,W));

MAX_CAP(N,R,T,W)$ (conec(N,R,T,W) EQ 1).. f(N,R,T,W) =l= fmax(N,R);

Pgenerated(I,T,W)$ (G(I,T,W) EQ 1)..

Pg(I,T,W) =e= Ps(I,T)+ r_spin_depG_u(I,T,W)+r_nspin_depG(I,T,W)
-r_spin_depG_d(I,T,W);

Pconsumed(J,T,W)..

Lc(J,T,W) =e= Ls(J,T)- r_spin_depD_u(J,T,W) + r_spin_depD_d(J,T,W);

RESERVE_DPL_SPIN_G_u(I,T,W).. r_spin_depG_u(I,T,W) =l= R_spin_schG_u(I,T);

```

```

RESERVE_DPL_SPIN_G_d(I,T,W).. r_spin_depG_d(I,T,W) =l= R_spin_schG_d(I,T);

RESERVE_DPL_NSPIN_G(I,T,W).. r_nspin_depG(I,T,W) =l= R_nspin_schG(I,T);

RESERVE_DPL_SPIN_D_u(J,T,W).. r_spin_depD_u(J,T,W)=l= R_spin_schD_u(J,T);

RESERVE_DPL_SPIN_D_d(J,T,W).. r_spin_depD_d(J,T,W)=l= R_spin_schD_d(J,T);

*****
*      MODEL
*****

MODEL MCU /ALL/;
MCU.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option mip=cplex;
SOLVE MCU USING mip MINIMIZING EC;

```

A.9 GAMS code for the Market-Clearing Example with Wind Generation (Section 11.5)

```

$title MARKET CLEARING UNDER WIND GENERATION UNCERTAINTY

*****
*      DATA
*****
SETS
N      Nodes                /n1 * n3/
I      Units                /i1 * i3/
J      Loads                /j1 * j1/
T      Periods              /t1 * t4/
W      Wind power scenarios /w1 * w3/
M      Offer blocks         /m1 * m1/;

ALIAS(N,R);

SCALARS
NT      Number of time periods /4/
dt      Duration of time periods /1/
x      Reactance of transmission lines /0.13/
Pbase   Power base /41/
wind_bus System node at which wind generation is injected /2/;

TABLE GDATA(I,*) Generation technical data
      PMIN   PMAX   Istatus   Pini
i1    10    100    0         0
i2    10    100    0         0
i3    10    50    0         0;

TABLE RDATA_G(I,T,*) Generation-side reserve data
      Rspin_max_up   Rspin_max_dn   Rnspin_max
i1.t1*t4           90             90         100
i2.t1*t4           90             90         100
i3.t1*t4           40             40         50;

TABLE CostBlock(I,T,M) Offer cost of energy blocks

```

```

                                m1
i1.t1*t4      30
i2.t1*t4      40
i3.t1*t4      20;

TABLE WidthBlock(I,T,M) Width of energy blocks
                                m1
i1.t1*t4      100
i2.t1*t4      100
i3.t1*t4      50;

TABLE Cr_schG(I,T,*) Generation-side reserve offer cost
                                up      dn      NS
i1.t1*t4      5      5      4.5
i2.t1*t4      7      7      5.5
i3.t1*t4      8      8      7;

TABLE ML(J,N) Mapping of the set of loads into the set of nodes
                                n1      n2      n3
j1      0      0      1;

TABLE MG(I,N) Mapping of the set of units into the set of nodes
                                n1      n2      n3
i1      1      0      0
i2      0      1      0
i3      0      0      1;

TABLE WP(T,W) Scenarios of wind power production
                                w1      w2      w3
t1      6      9      2
t2      20     30     13
t3      35     50     25
t4      8      12     6;

PARAMETERS
elast_limits(J,T,*) Elasticity limits of loads
/  j1.t1.sup      30
   j1.t2.sup      80
   j1.t3.sup      110
   j1.t4.sup      40
   j1.t1.inf      30
   j1.t2.inf      80
   j1.t3.inf      110
   j1.t4.inf      40 /

RDATA_D(J,T,*) Demand-side reserve data
/  j1.t1.Rspin_max_up      3
   j1.t2.Rspin_max_up      8
   j1.t3.Rspin_max_up      11
   j1.t4.Rspin_max_up      4
   j1.t1.Rspin_max_dn      3
   j1.t2.Rspin_max_dn      8
   j1.t3.Rspin_max_dn      11
   j1.t4.Rspin_max_dn      4 /

prob(W) Scenario w probability
/  w1      0.6
   w2      0.2
   w3      0.2 /

Num0(I,T)      Number of energy blocks offered by each unit
lambdaSU(I,T)  Start-up offer cost
lambdaL(J,T)   Demand utility
Cr_schD(J,T,*) Demand-side reserve offer cost
Vlol(J,T)      Value of lost load
Vspill(T)      Cost of wind power spillage
conec(N,N)     Network topology

```

```

b(N,N)          Imaginary parts of the admittance of lines
fmax(N,N)       Transmission capacity limits;

NumO(I,T) = 1;
lambdaSU(I,T) = 100;
lambdaL(J,T) = 0;
Cr_schD(J,T,'up') = 70;
Cr_schD(J,T,'dn') = 70;
Vlol(J,T) = 1000;
Vspill(T) = 0;
conec(N,R)$(ord(N) ne ord(R)) = 1;
b(N,R)$(ord(N) ne ord(R)) = -1/x;
fmax(N,R)$(ord(N) ne ord(R)) = 55;

*****
*      DECLARATION OF VARIABLES
*****

VARIABLES

    EC                      Expected cost (objective function value)

** FIRST-STAGE VARIABLES

    Ps(I,T)                Power output schedule
    Ls(J,T)                Demand schedule
    WPs(T)                 Wind power schedule
    R_spin_schG_d(I,T)     Generation-side spinning reserve schedule (downward)
    R_spin_schG_u(I,T)     Generation-side spinning reserve schedule (upward)
    R_spin_schD_d(J,T)     Demand-side spinning reserve schedule (downward)
    R_spin_schD_u(J,T)     Demand-side spinning reserve schedule (upward)
    R_nspin_schG(I,T)      Generation-side non-spinning reserve schedule
    u(I,T)                 Scheduled commitment status

** SECOND-STAGE VARIABLES

    Lc(J,T,W)              Power consumption in real time
    Lshed(J,T,W)           Amount of involuntarily load shed
    s(T,W)                 Amount of spilled wind power production
    Pg(I,T,W)              Generator power output in real time
    pgblock(I,T,W,M)       Power produced from energy blocks
    r_spin_depG_d(I,T,W)   Generation-side spinning reserve deployment (downward)
    r_spin_depG_u(I,T,W)   Generation-side spinning reserve deployment (upward)
    r_spin_depD_d(J,T,W)   Demand-side spinning reserve deployment (downward)
    r_spin_depD_u(J,T,W)   Demand-side spinning reserve deployment (upward)
    r_nspin_depG(I,T,W)    Generation-side non-spinning reserve deployment
    angle(N,T,W)           Voltage angle
    Pinj(N,T,W)            Power injection
    f(N,R,T,W)             Power flow
    v(I,T,W)              Real-time commitment status
    Csu(I,T,W)             Start-up cost;

*****
*      MATHEMATICAL CHARACTERIZATION OF VARIABLES
*****

POSITIVE VARIABLES

    Csu(I,T,W), Lshed(J,T,W), r_spin_depG_d(I,T,W), r_spin_depG_u(I,T,W),
    r_nspin_depG(I,T,W), r_spin_depD_d(J,T,W), r_spin_depD_u(J,T,W),
    R_spin_schG_d(I,T), R_spin_schG_u(I,T), R_spin_schD_d(J,T), R_spin_schD_u(J,T),
    R_nspin_schG(I,T), s(T,W), pgblock(I,T,W,M), WPs(T);

BINARY VARIABLES u(I,T), v(I,T,W);

** Elasticity limits of demand

```

```

Ls.up(J,T)= elast_limits(J,T,'sup');
Ls.lo(J,T)= elast_limits(J,T,'inf');

** Reference node

angle.fx('N1',T,W)=0;

** Wind generation limits

WPs.up(T)= 60;

*****
*      EQUATIONS
*****

EQUATIONS

ECFunction      Objective function

** ELECTRICITY MARKET CONSTRAINTS

** Production limits

MAX_PROD(I,T)
MIN_PROD(I,T)

** Market equilibria

MARKET_EQU(T)

** Scheduled reserve determination constraints

** Generation side

RESERVE_SCH_SPIN_G_u(I,T) Spinning up
RESERVE_SCH_SPIN_G_d(I,T) Spinning down
RESERVE_SCH_NSPIN_G(I,T)  Non-spinning

** Demand side

RESERVE_SCH_SPIN_D_u(J,T) Spinning up
RESERVE_SCH_SPIN_D_d(J,T) Spinning down

** REAL-TIME OPERATING CONSTRAINTS

** Start-up costs

SUo(I,W)      T = 1
SU(I,T,W)     T > 1

** Generation limits

GL1(I,T,W)
GL2(I,T,W)

** Decomposition of generator power outputs into blocks

GL3(I,T,W,M)
GL4(I,T,W)

** Involuntary load shedding constraints

LIMIT_ENS(J,T,W)

** Limit of wind power generation spillage

```

```

LIMIT_SPILLAGE(T,W)

** Network constraints

PB(N,T,W)           Power balance
PB_wind_bus(N,T,W)  Power balance at wind bus
DEF_Pinj(N,T,W)     Definition of power injection
DEF_FLOW(N,R,T,W)   Definition of power flows
MAX_CAP(N,R,T,W)    Transmission capacity

** LINKING CONSTRAINTS

Pgenerated(I,T,W)    Composition of generator power outputs
Pconsumed(J,T,W)     Composition of the power consumption

** Deployed reserve determination constraints

** Generation side

RESERVE_DPL_SPIN_G_u(I,T,W) Spinning up
RESERVE_DPL_SPIN_G_d(I,T,W) Spinning down
RESERVE_DPL_NSPIN_G(I,T,W)  Non-spinning

** Demand side

RESERVE_DPL_SPIN_D_u(J,T,W) Spinning up
RESERVE_DPL_SPIN_D_d(J,T,W) Spinning down
;

ECFunction..
EC =e= dt*sum((I,T),
Cr_schG(I,T,'up')* R_spin_schG_u(I,T)+ Cr_schG(I,T,'dn')* R_spin_schG_d(I,T)
+ Cr_schG(I,T,'NS')* R_nspin_schG(I,T)) + dt*sum((J,T),
Cr_schD(J,T,'up')* R_spin_schD_u(J,T)+ Cr_schD(J,T,'dn')* R_spin_schD_d(J,T))
+ dt*sum((T,W),
prob(W)*(sum((I,M)$ (ord(M) LE Num0(I,T)), CostBlock(I,T,M)* pgblock(I,T,W,M))
- sum(J, lambdaL(J,T)* Lc(J,T,W))
+ sum(J, Vlo1(J,T)*Lshed(J,T,W)) + Vspill(T)*s(T,W)))
+ sum((I,T,W),prob(W)*Csu(I,T,W)));

MIN_PROD(I,T).. Ps(I,T) =g= GDATA(I,'PMIN')* u(I,T);

MAX_PROD(I,T).. Ps(I,T) =l= GDATA(I,'PMAX')*u(I,T);

MARKET_EQU(T).. sum(I, Ps(I,T)) + WPs(T) =e= sum(J, Ls(J,T));

RESERVE_SCH_SPIN_G_u(I,T)..
R_spin_schG_u(I,T) =l= RDATA_G(I,T,'Rspin_max_up')*u(I,T);

RESERVE_SCH_SPIN_G_d(I,T)..
R_spin_schG_d(I,T) =l= RDATA_G(I,T,'Rspin_max_dn')*u(I,T);

RESERVE_SCH_NSPIN_G(I,T)..
R_nspin_schG(I,T) =l= RDATA_G(I,T,'Rnspin_max')*(1-u(I,T));

RESERVE_SCH_SPIN_D_u(J,T)..
R_spin_schD_u(J,T) =l= RDATA_D(J,T,'Rspin_max_up');

RESERVE_SCH_SPIN_D_d(J,T)..
R_spin_schD_d(J,T) =l= RDATA_D(J,T,'Rspin_max_dn');

SUo(I,W).. Csu(I,'t1',W) =g= lambdaSU(I,'t1')*(v(I,'t1',W)-GDATA(I,'Istatus'));

SU(I,T,W)$ (ord(T) GT 1).. Csu(I,T,W) =g= lambdaSU(I,T)*(v(I,T,W)-v(I,T-1,W));

GL1(I,T,W).. Pg(I,T,W) =g= GDATA(I,'PMIN')* v(I,T,W);

```

```

GL2(I,T,W).. Pg(I,T,W) =l= GDATA(I,'PMAX')* v(I,T,W);

GL3(I,T,W,M)$ (ord(M) LE NumO(I,T)).. pgblock(I,T,W,M) =l= WidthBlock(I,T,M);

GL4(I,T,W).. Pg(I,T,W) =e= sum(M$(ord(M) LE NumO(I,T)), pgblock(I,T,W,M));

LIMIT_ENS(J,T,W).. Lshed(J,T,W) =l= Lc(J,T,W);

LIMIT_SPILLAGE(T,W).. s(T,W) =l= WP(T,W);

PB(N,T,W)$ (ord(N) NE wind_bus)..

sum(I$(MG(I,N) EQ 1),Pg(I,T,W))- sum(J$(ML(J,N) EQ 1),
    Lc(J,T,W)-Lshed(J,T,W))-Pinj(N,T,W) =e= 0;

PB_wind_bus(N,T,W)$ (ord(N) EQ wind_bus)..
sum(I$(MG(I,N) EQ 1),Pg(I,T,W))- sum(J$(ML(J,N) EQ 1), Lc(J,T,W)-Lshed(J,T,W))
    + WP(T,W)-s(T,W)-Pinj(N,T,W) =e= 0;

DEF_Pinj(N,T,W)..
    Pinj(N,T,W) =e= sum(R$(conec(N,R) EQ 1),f(N,R,T,W));

DEF_FLOW(N,R,T,W)$ (conec(N,R) EQ 1)..
    f(N,R,T,W) =e= -b(N,R)*(angle(N,T,W)-angle(R,T,W));

MAX_CAP(N,R,T,W)$ (conec(N,R) EQ 1)..
    f(N,R,T,W) =l= fmax(N,R);

Pgenerated(I,T,W)..
    Pg(I,T,W) =e= Ps(I,T)+ r_spin_depG_u(I,T,W)+r_nspin_depG(I,T,W)
        -r_spin_depG_d(I,T,W);

Pconsumed(J,T,W)..
    Lc(J,T,W) =e= Ls(J,T)- r_spin_depD_u(J,T,W) + r_spin_depD_d(J,T,W);

RESERVE_DPL_SPIN_G_u(I,T,W)..
    r_spin_depG_u(I,T,W) =l= R_spin_schG_u(I,T);

RESERVE_DPL_SPIN_G_d(I,T,W)..
    r_spin_depG_d(I,T,W) =l= R_spin_schG_d(I,T);

RESERVE_DPL_NSPIN_G(I,T,W)..
    r_nspin_depG(I,T,W) =l= R_nspin_schG(I,T);

RESERVE_DPL_SPIN_D_u(J,T,W)..
    r_spin_depD_u(J,T,W)=l= R_spin_schD_u(J,T);

RESERVE_DPL_SPIN_D_d(J,T,W)..
    r_spin_depD_d(J,T,W)=l= R_spin_schD_d(J,T);

*****
*           MODEL
*****

MODEL MCU_WP /ALL/;
MCU_WP.optcr=0;
Option iterlim = 1e8;
Option reslim = 1e10;
Option mip=cplex;
SOLVE MCU_WP USING mip MINIMIZING EC;

```

Appendix B

24-Node System Data

This appendix lists the characteristics of the 24-node system used to examine the market-clearing models described in Chapters 10 and 11. Unless stated otherwise in the main text of these chapters, the characteristics itemized next apply integrally.

B.1 Network data

The 24-node system considered in this book is based on the single-area version of the IEEE Reliability Test System–1996 [54]. The system comprises 34 transmission lines as illustrated in Fig. B.1. Line resistances are null and thus, active power losses are disregarded. The values of reactance and capacity of transmission lines are listed in Table B.1. Line reactances are given in per unit on a 100-MVA base.

B.2 Generator data

Technical data for the generators are given in Table B.2. The power system includes 12 generating units. Unit 10 is a hydro generator whereas the rest of them are all thermal. In particular, units 8 and 9 are nuclear power plants. Table B.3 indicates the node location of each unit. The nuclear and hydro generators are assumed to be must-run units.

For simplicity, the energy and reserve offers of the units are assumed to be the same throughout the scheduling horizon consisting of 24 hourly periods. Tables B.4 and B.5 provide, respectively, the blocks of energy submitted by each generating unit and their corresponding offer costs. Table B.6 lists the start-up cost *declared* by each unit. Each generator offers the maximum possible amount of up and down-spinning reserve ($P^{\max} - P^{\min}$), both at

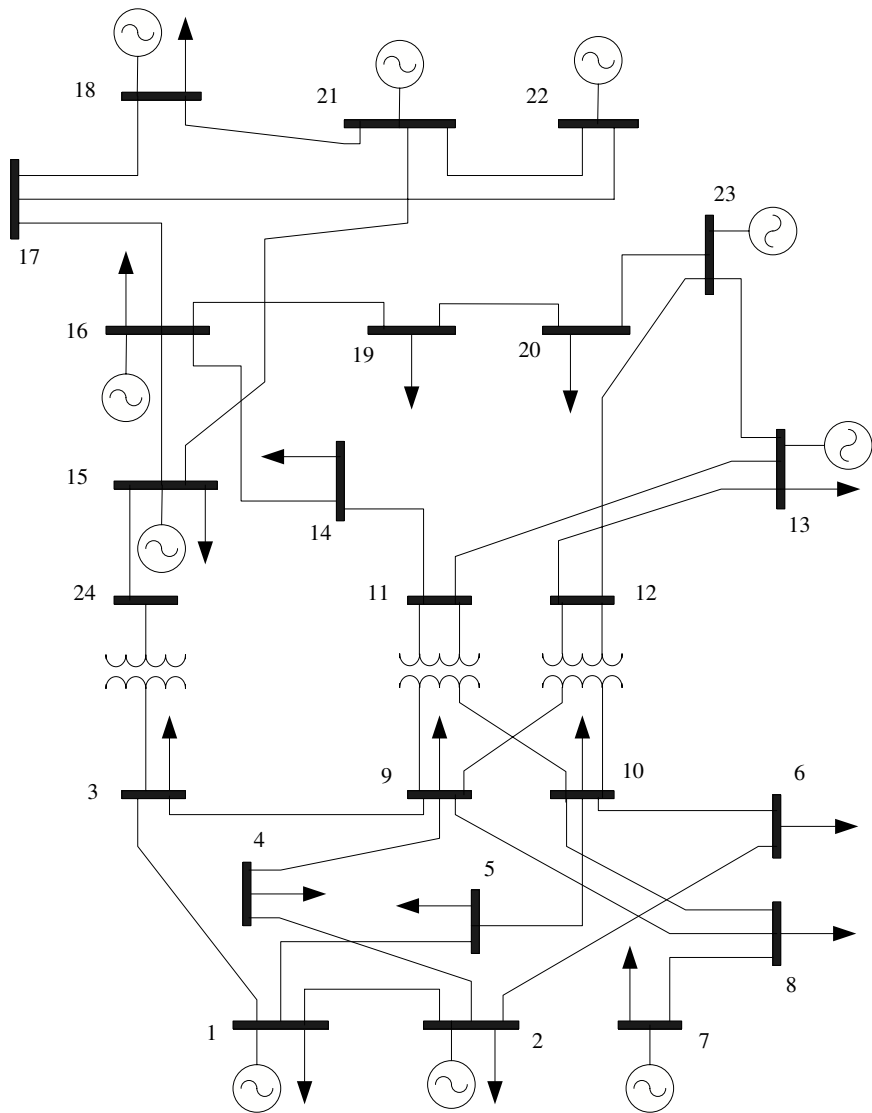


Fig. B.1 24-node system

Table B.1 24-node system: reactance and capacity of transmission lines

From node	To node	Reactance (p.u.)	Capacity (MVA)
1	2	0.0146	175
1	3	0.2253	175
1	5	0.0907	175
2	4	0.1356	175
2	6	0.2050	175
3	9	0.1271	175
3	24	0.0840	400
4	9	0.1110	175
5	10	0.0940	175
6	10	0.0642	175
7	8	0.0652	175
8	9	0.1762	175
8	10	0.1762	175
9	11	0.0840	400
9	12	0.0840	400
10	11	0.0840	400
10	12	0.0840	400
11	13	0.0488	500
11	14	0.0426	500
12	13	0.0488	500
12	23	0.0985	500
13	23	0.0884	500
14	16	0.0594	500
15	16	0.0172	500
15	21	0.0249	1000
15	24	0.0529	500
16	17	0.0263	500
16	19	0.0234	500
17	18	0.0143	500
17	22	0.1069	500
18	21	0.0132	1000
19	20	0.0203	1000
20	23	0.0112	1000
21	22	0.0692	500

Table B.2 24-node system: technical data of generating units

Unit #	1	2	3	4	5	6	7	8	9	10	11	12
$P^{\min}$ (MW)	30.4	30.4	75	206.85	12	54.25	54.25	100	100	300	108.5	140
$P^{\max}$ (MW)	152	152	300	591	60	155	155	400	400	300	310	350
RD (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
RU (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
SD (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
SU (MW/h)	152	152	300	540	60	155	155	400	400	300	310	240
D^T (h)	4	4	8	10	2	8	8	1	1	0	8	48
U^T (h)	8	8	8	12	4	8	8	1	1	0	8	24
S^0 (h)	0	0	2	1	1	2	0	0	0	0	0	0
V^0 (h)	22	22	0	0	0	0	10	769	16	24	10	300
u^0 (h)	1	1	0	0	0	0	1	1	1	1	1	1

$P^{\min}$: minimum power output; $P^{\max}$: maximum power output; RD: ramp-down limit; RU: ramp-up limit; SD: shut-down ramp limit; SU: start-up ramp limit; D^T : minimum down time; U^T : minimum up time; S^0 : time periods the generating unit has been offline at the beginning of the scheduling horizon; V^0 : time periods the generating unit has been online at the beginning of the scheduling horizon; u^0 : initial commitment status (1 online, 0 otherwise)

Table B.3 24-node system: node location of generating units

Unit #	1	2	3	4	5	6	7	8	9	10	11	12
Node location	1	2	7	13	15	15	16	18	21	22	23	23

a cost equal to 25% of its highest offer cost of energy production. However, only generators located at buses 7, 15 and 16 offer non-spinning reserve. We consider that these units constitute the set of generators with the required features to provide this reserve service. Then, each of these generating units offers the maximum possible amount of non-spinning reserve ($P^{\max}$) at a cost equal to 20% of its highest offer cost of energy production.

B.3 Demand data

The load profile for the considered 24-hour scheduling horizon is provided in Table B.7. The 24-node system sketched in Fig. B.1 includes 17 loads. Both the nodal location of these loads and their contribution in percentage to the total system demand are listed in Table B.8. All these loads offer identical rates of \$50/MWh for up and down-spinning reserves. Loads can be curtailed from their normal levels up to 2% of their scheduled consumption for all hours to provide up-spinning reserves. Likewise, the demands can increase the same

Table B.4 24-node system: size (MWh) of the energy blocks offered by generating units

Unit #	Block #			
	1	2	3	4
1	30.40	45.60	45.60	30.40
2	30.40	45.60	45.60	30.40
3	75.00	75.00	90.00	60.00
4	206.85	147.75	118.20	118.20
5	12.00	18.00	18.00	12.00
6	54.25	38.75	31.00	31.00
7	54.25	38.75	31.00	31.00
8	100.00	100.00	120.00	80.00
9	100.00	100.00	120.00	80.00
10	300.00	0.00	0.00	0.00
11	108.50	77.50	62.00	62.00
12	140.00	87.50	52.50	70.00

Table B.5 24-node system: offer costs (\$/MWh) submitted by generating units

Unit #	Block #			
	1	2	3	4
1	11.46	11.96	13.89	15.97
2	11.46	11.96	13.89	15.97
3	18.60	20.03	21.67	22.72
4	19.20	20.32	21.22	22.13
5	23.41	23.78	26.84	30.40
6	9.92	10.25	10.68	11.26
7	9.92	10.25	10.68	11.26
8	5.31	5.38	5.53	5.66
9	5.31	5.38	5.53	5.66
10	0.00	0.00	0.00	0.00
11	9.92	10.25	10.68	11.26
12	10.08	10.66	11.09	11.72

quantity to contribute to down-spinning reserve. Beyond these limits, the value of lost load is assumed to be \$2000/MWh.

Table B.6 24-node system: start-up costs of generating units

Unit #	Start-up cost (\$)	Unit #	Start-up cost (\$)
1	1430.4	7	312.0
2	1430.4	8	0.0
3	1725.0	9	0.0
4	3056.7	10	0.0
5	437.0	11	624.0
6	312.0	12	2298.0

Table B.7 24-node system: load profile

Hour	System demand (MW)	Hour	System demand (MW)
12-1 am	1775.835	noon-1 pm	2517.975
1-2	1669.815	1-2	2517.975
2-3	1590.300	2-3	2464.965
3-4	1563.795	3-4	2464.965
4-5	1563.795	4-5	2623.995
5-6	1590.300	5-6	2650.500
6-7	1961.370	6-7	2650.500
7-8	2279.430	7-8	2544.480
8-9	2517.975	8-9	2411.955
9-10	2544.480	9-10	2199.915
10-11	2544.480	10-11	1934.865
11-noon	2517.975	11-12	1669.815

Table B.8 24-node system: node location and distribution of the total system demand

Load #	Node #	% of system load	Load #	Node #	% of system load
1	1	3.8	10	10	6.8
2	2	3.4	11	13	9.3
3	3	6.3	12	14	6.8
4	4	2.6	13	15	11.1
5	5	2.5	14	16	3.5
6	6	4.8	15	18	11.7
7	7	4.4	16	19	6.4
8	8	6.0	17	20	4.5
9	9	6.1			

Appendix C

Exercise solutions

This chapter includes the solutions to some of the exercises proposed in Chapters 2 to 11. These exercises are computationally oriented and serve as a complement to the illustrative examples solved throughout these chapters. Exercises from Chapter 1 are not solved due to their conceptual nature. The purpose of this appendix is to help the reader to further comprehend the nature of the models and solutions techniques described in this book, as well as to develop appropriate skills to formulate and solve similar models.

C.1 Exercises from Chapter 2

Solution to Exercise 2.3

An electricity producer owns a 100-MW unit with a production cost of \$25/MWh, and faces an uncertain electricity selling price for the next week. For simplicity, we consider that the price is uncertain but constant throughout the week. Scenario price data are provided in Table C.1.

Table C.1 Solution to Exercise 2.3: scenario data for the producer

Scenario #	Probability (per unit)	Price (\$/MWh)
1	0.2	50
2	0.6	46
3	0.2	44

Additionally, this producer has the possibility of selling up to 60 MW at \$45/MWh over the next week, by signing a bilateral contract before that week, i.e., prior to knowing the actual pool price it has to face.

The two-stage stochastic programming problem that model the decision-making process of the producer is as follows:

$$\begin{aligned}
 & \text{Maximize}_{P_1^C, P_2^C, P_3^C, P_1, P_2, P_3} \\
 & \Pi^S = \\
 & 168(0.2 \times 45P_1^C + 0.6 \times 45P_2^C + 0.2 \times 45P_3^C) + \\
 & 168(0.2 \times 50P_1 + 0.6 \times 46P_2 + 0.2 \times 44P_3) - \\
 & 168 \times 25 [0.2 \times (P_1^C + P_1) + 0.6 \times (P_2^C + P_2) + 0.2 \times (P_3^C + P_3)] \\
 & \text{subject to} \\
 & P_1^C + P_1 \leq 100 \\
 & P_2^C + P_2 \leq 100 \\
 & P_3^C + P_3 \leq 100 \\
 & 0 \leq P_1^C, P_2^C, P_3^C \leq 60 \\
 & 0 \leq P_1, P_2, P_3 \\
 & P_1^C = P_2^C = P_3^C .
 \end{aligned}$$

Variables P_1^C , P_2^C and P_3^C represent the power sold through the bilateral contract in scenarios 1, 2 and 3, respectively. They should be equal so that no information is anticipated (as enforced by the last constraint). Variables P_1 , P_2 , and P_3 represent the power sold in the pool for scenarios 1, 2, and 3, respectively.

The objective function is the expected profit of the producer: revenue from selling through the bilateral contract (first row of the objective function) plus revenue from selling in the pool (second row) minus production cost (third row). Powers are multiplied by 168 to obtain the energy produced throughout the week.

The first three constraints enforce the capacity limit of the production unit for the three scenarios, while the next two constraints are the contract bound and nonnegativity declarations for all the variables. The last constraint enforces the non-anticipativity of the information.

The solution to this problem is $P_1^{C*} = P_2^{C*} = P_3^{C*} = 0$, $P_1^* = P_2^* = P_3^* = 100$ MW, which means that the power producer decides to sell its full production capacity to the pool regardless of the eventual pool price realization. This is so because the expected value of the pool price during the week, \$46.4/MWh, is greater than the price of the available bilateral contract, \$45/MWh.

The optimal expected profit is $\Pi^{S*} = \$359,520$.

Solution to Exercise 2.4

This exercise is built upon the solution to Exercise 2.3 above. The decision-making process of the producer of Exercise 2.3 is as follows:

- a. At the first stage (root of the tree in Fig. C.1), the producer has to decide how much to sell through the contract. This is modeled through variables $P_1^C = P_2^C = P_3^C = P^C$.
- b. The second stage reproduces pool selling for each of the three considered price scenarios (leaves of the tree in Fig. C.1). This is modeled through variables P_1 , P_2 and P_3 .

The corresponding scenario tree of the producer decision-making process is provided in Fig. C.1.

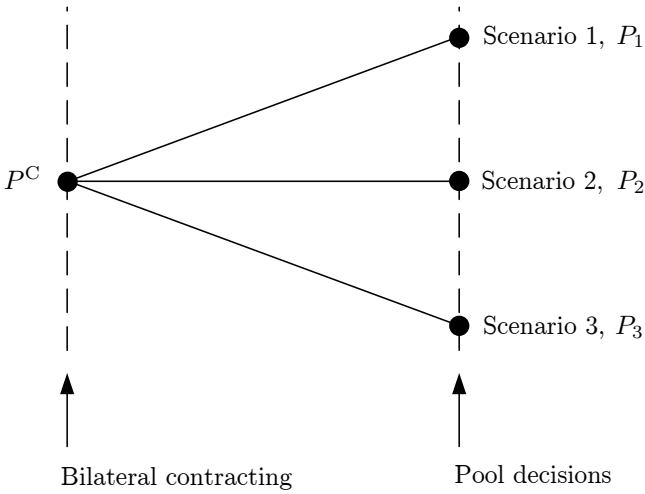


Fig. C.1 Solution to Exercise 2.4: scenario tree for the producer

Solution to Exercise 2.5

This exercise is build upon the solution to Exercises 2.3 and 2.4 above. EVPI is computed as follows. First, the problem below is solved,

$$\begin{aligned}
& \text{Maximize}_{P_1^C, P_2^C, P_3^C, P_1, P_2, P_3} \\
& \Pi^P = \\
& 168(0.2 \times 45P_1^C + 0.6 \times 45P_2^C + 0.2 \times 45P_3^C) + \\
& 168(0.2 \times 50P_1 + 0.6 \times 46P_2 + 0.2 \times 44P_3) - \\
& 168 \times 25 [0.2 \times (P_1^C + P_1) + 0.6 \times (P_2^C + P_2) + 0.2 \times (P_3^C + P_3)] \\
& \text{subject to} \\
& P_1^C + P_1 \leq 100 \\
& P_2^C + P_2 \leq 100 \\
& P_3^C + P_3 \leq 100 \\
& 0 \leq P_1^C, P_2^C, P_3^C \leq 60 \\
& 0 \leq P_1, P_2, P_3.
\end{aligned}$$

Note that in this problem a contracting variable is defined per scenario, which implies perfect information. In other words, the contracting decision is made dependent on the pool price scenario. This problem is similar to that of Exercise 2.3 but eliminating the last constraint (non-anticipativity constraint). Its optimal objective function value is $\Pi^{P*} = \$361,536$.

Then, EPVI is calculated as

$$\text{EPVI} = \Pi^{P*} - \Pi^{S*} = \$361,536 - \$359,520 = \$2016,$$

where Π^{S*} is the objective function optimal value of the problem in Exercise 2.3 above. Therefore, the error associated with the pool price forecast is expected to entail a cost of \$2016 to the power producer.

Next, VSS is computed as follows. Since the average price value is $(0.2 \times 50 + 0.6 \times 46 + 0.2 \times 44) = \$46.4/\text{MWh}$, the deterministic version of the problem in Exercise 2.3 has the form

$$\begin{aligned}
& \text{Maximize}_{P^C, P} \\
& 168 [45P^C + 46.4P - 25(P + P^C)] \\
& \text{subject to} \\
& P^C + P \leq 100 \\
& 0 \leq P^C \leq 60 \\
& 0 \leq P,
\end{aligned}$$

where P^C is the power sold through the bilateral contract and P the power sold in the pool. The solution to this problem is $P^{C*} = 0$ MW and $P^* = 100$ MW, implying a profit of \$359,520.

Then, we solve the stochastic problem of Exercise 2.3 but fixing P^C to its optimal value obtained from solving the deterministic problem above, i.e., to 0,

$$\begin{aligned}
& \text{Maximize}_{P_1^C, P_2^C, P_3^C, P_1, P_2, P_3} \\
& \Pi^D = \\
& 168(0.2 \times 45P_1^C + 0.6 \times 45P_2^C + 0.2 \times 45P_3^C) + \\
& 168(0.2 \times 50P_1 + 0.6 \times 46P_2 + 0.2 \times 44P_3) - \\
& 168 \times 25 [0.2 \times (P_1^C + P_1) + 0.6 \times (P_2^C + P_2) + 0.2 \times (P_3^C + P_3)] \\
& \text{subject to} \\
& P_1^C + P_1 \leq 100 \\
& P_2^C + P_2 \leq 100 \\
& P_3^C + P_3 \leq 100 \\
& 0 \leq P_1^C, P_2^C, P_3^C \leq 60 \\
& 0 \leq P_1, P_2, P_3 \\
& P_1^C = P_2^C = P_3^C = 0,
\end{aligned}$$

whose solution is $P_1^* = P_2^* = P_3^* = 100$ MW and $\Pi^{D*} = \$359,520$,
VSS is then computed as

$$\text{VSS} = \Pi^{S*} - \Pi^{D*} = \$359,520 - \$359,520 = \$0,$$

where Π^S is the objective function optimal value resulting from the stochastic version of the producer problem described in Exercise 2.3. Consequently, no economic advantage is obtained from using a stochastic approach to solve this problem over a deterministic one. In the solution to Exercise 6.7, the reader can find an example of a problem in which the VSS is different from zero.

C.2 Exercises from Chapter 3

Solution to Exercise 3.1

We denote by $y_{t\omega}$ the realization of process y_t in scenario ω . For this process, the ARMA(2,0,1) model is

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} - \theta_1 \epsilon_{t-1} + \epsilon_t, \quad (\text{C.1})$$

that particularized for scenario ω results in

$$y_{t\omega} = \phi_1 y_{t-1\omega} + \phi_2 y_{t-2\omega} - \theta_1 \epsilon_{t-1\omega} + \epsilon_{t\omega}. \quad (\text{C.2})$$

For each period and scenario, the scenario-generation procedure consists in randomly generating a value of the white noise term $\epsilon_{t\omega}$ and evaluating expression (C.2) to obtain $y_{t\omega}$.

Being $y_{0\omega} = 125.2134$, $y_{1\omega} = 120.1054$, and $\epsilon_{0\omega} = 0.0124$ (for all ω), the procedure used to obtain three four-period scenarios is the following:

Scenario 1.

Period 1: $\epsilon_{11} = 1.5861$.

$$y_{11} = 0.8124 \times 125.2134 + 0.2514 \times 120.1054 - 0.8714 \times 0.0124 + 1.5861 = 131.3429.$$

Period 2: $\epsilon_{21} = 2.0336$.

$$y_{21} = 0.8124 \times 131.3429 + 0.2514 \times 125.2134 - 0.8714 \times 1.5861 + 2.0336 = 138.8331.$$

Period 3: $\epsilon_{31} = 2.3946$.

$$y_{31} = 0.8124 \times 138.8331 + 0.2514 \times 131.3429 - 0.8714 \times 2.0336 + 2.3946 = 146.4301.$$

Period 4: $\epsilon_{41} = -2.5249$.

$$y_{41} = 0.8124 \times 146.4301 + 0.2514 \times 138.8331 - 0.8714 \times 2.3946 - 2.5249 = 149.2509.$$

Scenario 2.

Period 1: $\epsilon_{12} = 0.5396$.

$$y_{12} = 0.8124 \times 125.2134 + 0.2514 \times 120.1054 - 0.8714 \times 0.0124 + 0.5396 = 130.2964.$$

Period 2: $\epsilon_{22} = 0.6054$.

$$y_{22} = 0.8124 \times 130.2964 + 0.2514 \times 125.2134 - 0.8714 \times 0.5396 + 0.6054 = 137.4666.$$

Period 3: $\epsilon_{32} = -2.5648$.

$$y_{32} = 0.8124 \times 137.4666 + 0.2514 \times 130.2964 - 0.8714 \times 0.6054 - 2.5648 = 141.3421.$$

Period 4: $\epsilon_{42} = -1.8885$.

$$y_{42} = 0.8124 \times 141.3421 + 0.2514 \times 137.4666 + 0.8714 \times 2.5648 - 1.8885 = 149.7319.$$

Scenario 3.

Period 1: $\epsilon_{13} = 2.7544$.

$$y_{13} = 0.8124 \times 125.2134 + 0.2514 \times 120.1054 - 0.8714 \times 0.0124 + 2.7544 = 132.5112.$$

Period 2: $\epsilon_{23} = -0.3347$.

$$y_{23} = 0.8124 \times 132.5112 + 0.2514 \times 125.2134 - 0.8714 \times 2.7544 - 0.3347 = 136.3959.$$

Period 3: $\epsilon_{33} = 0.9922$.

$$y_{33} = 0.8124 \times 136.3959 + 0.2514 \times 132.5112 + 0.8714 \times 0.3347 + 0.9922 = 145.4052.$$

Period 4: $\epsilon_{43} = 0.2239$.

$$y_{43} = 0.8124 \times 145.4052 + 0.2514 \times 136.3959 - 0.8714 \times 0.9922 + 0.2239 = 151.7764.$$

The resulting set of scenarios of process y_t is depicted in Fig. C.2.

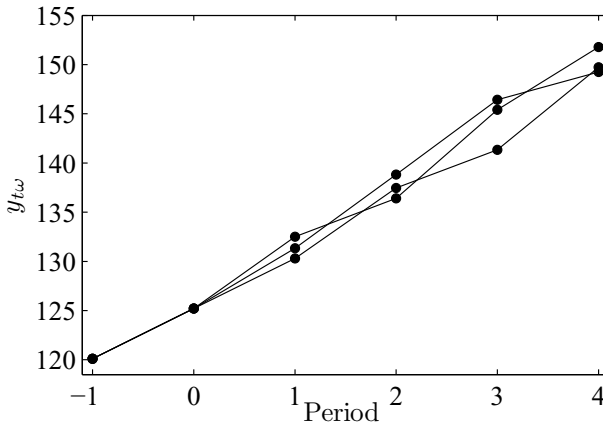


Fig. C.2 Solution to Exercise 3.1: scenarios of y_t

Solution to Exercise 3.4

We follow the algorithm provided in Subsection 3.2.3 of Chapter 3.

Scenario 1.

Iteration 1: We randomly generate $u_1, u_2 \sim U(0, 1)$, where $u_1 = 0.9388$ and $u_2 = 0.9961$.

The time to failure (t_f) and the time to repair (t_r) are obtained using the randomly generated values of u_1 and u_2 .

$$t_f = -2500 \times \ln(0.5383) = 157.9, \quad t_r = -150 \times \ln(0.9961) = 0.6.$$

The unit remains available throughout the first 158 hours, and it is off in hour 159.

For each scenario new values of t_f and t_r are successively obtained in order to fulfill the planning horizon (8760 hours).

Iteration 2: $u_1 = 0.0782$ and $u_2 = 0.4427$.

$$t_f = -2500 \times \ln(0.0782) = 6372, \quad t_r = -150 \times \ln(0.4427) = 122.2.$$

The unit remains available from hour 160 to hour 6531, and it is off from hour 6532 to hour 6653.

Iteration 3: $u_1 = 0.1067$ and $u_2 = 0.9619$.

$$t_f = -2500 \times \ln(0.1067) = 5595.4, \quad t_r = -150 \times \ln(0.9619) = 5.8.$$

The unit remains available from hour 6654 up to the end of the planning horizon (hour 8760).

Scenario 2.

Iteration 1: $u_1 = 0.0046$ and $u_2 = 0.7749$.

$$t_f = -2500 \times \ln(0.0046) = 13,454.2, \quad t_r = -150 \times \ln(0.7749) = 38.2.$$

The unit remains available the whole planning horizon (8760 hours).

Scenario 3.

Iteration 1: $u_1 = 0.8173$ and $u_2 = 0.8687$.

$$t_f = -2500 \times \ln(0.8173) = 504.3, \quad t_r = -150 \times \ln(0.8687) = 21.1.$$

The unit remains available the first 504 hours, and it is off from hour 505 to hour 525.

Iteration 2: $u_1 = 0.0844$ and $u_2 = 0.3998$.

$$t_f = -2500 \times \ln(0.0844) = 6179.4, \quad t_r = -150 \times \ln(0.3998) = 137.5.$$

The unit remains available from hour 526 to hour 6704, and it is off from hour 6705 to hour 6842.

Iteration 3: $u_1 = 0.2599$ and $u_2 = 0.8001$.

$$t_f = -2500 \times \ln(0.2599) = 3368.9, \quad t_r = -150 \times \ln(0.8001) = 33.4.$$

The unit remains available from hour 6843 up to the end of the planning horizon (hour 8760).

The resulting set of scenarios is depicted in Fig. C.3.

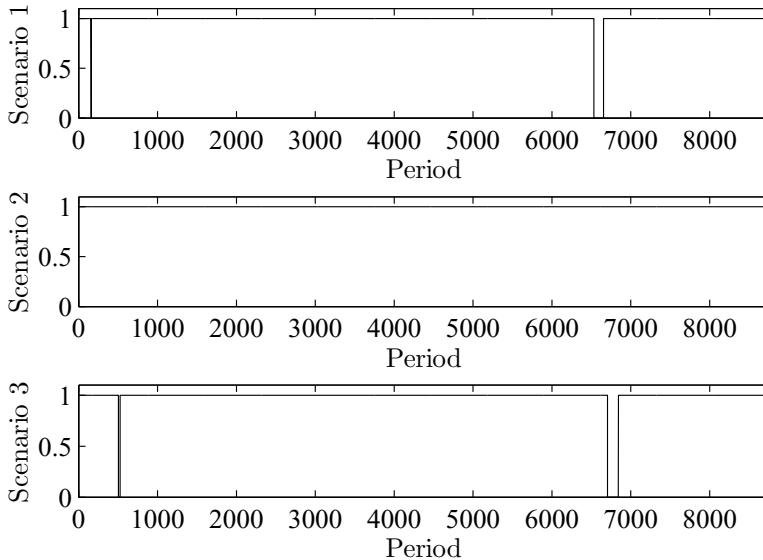


Fig. C.3 Solution to Exercise 3.4: availability scenarios

Solution to Exercise 3.6

The variant 1 of the fast forward selection algorithm described in Subsection 3.3.3 of Chapter 3 works as follows:

Step 0. We evaluate the cost function $\nu(\omega, \omega') = \|P_\omega - P_{\omega'}\|, \forall \omega, \omega' \in \Omega$, where $\Omega = \{1, 2, 3, 4, 5\}$. The values of function ν are arranged into a symmetric matrix with zero diagonal elements,

$$\nu = \begin{pmatrix} 0 & 5 & 22 & 30 & 35 \\ 5 & 0 & 17 & 25 & 30 \\ 22 & 17 & 0 & 8 & 13 \\ 30 & 25 & 8 & 0 & 5 \\ 35 & 30 & 13 & 5 & 0 \end{pmatrix} \text{ MW.}$$

Step 1. For each scenario ω that is candidate to be selected from Ω , we choose the one that minimizes the resulting Kantorovich distance between the reduced and original sets, Ω_S and Ω , respectively. Distance values are

$$\begin{aligned} d_1 &= \pi_2 \nu(1, 2) + \pi_3 \nu(1, 3) + \pi_4 \nu(1, 4) + \pi_5 \nu(1, 5) = 16.40 \text{ MW} \\ d_2 &= \pi_1 \nu(2, 1) + \pi_3 \nu(2, 3) + \pi_4 \nu(2, 4) + \pi_5 \nu(2, 5) = 13.90 \text{ MW} \\ d_3 &= \pi_1 \nu(3, 1) + \pi_2 \nu(3, 2) + \pi_4 \nu(3, 4) + \pi_5 \nu(3, 5) = 12.20 \text{ MW} \\ d_4 &= \pi_1 \nu(4, 1) + \pi_2 \nu(4, 2) + \pi_3 \nu(4, 3) + \pi_5 \nu(4, 5) = 14.60 \text{ MW} \\ d_5 &= \pi_1 \nu(5, 1) + \pi_2 \nu(5, 2) + \pi_3 \nu(5, 3) + \pi_4 \nu(5, 4) = 18.60 \text{ MW.} \end{aligned}$$

Therefore,

$$\begin{aligned} \Omega_S^{[1]} &= \{3\} \\ \Omega_J^{[1]} &= \{1, 2, 4, 5\}. \end{aligned}$$

Step 2. We update the cost matrix,

$$\begin{aligned} \nu^{[2]}(1, 2) &= \min\{\nu(1, 2), \nu(1, 3)\} = 5 \text{ MW} \\ \nu^{[2]}(1, 4) &= \min\{\nu(1, 4), \nu(1, 3)\} = 22 \text{ MW} \\ \nu^{[2]}(1, 5) &= \min\{\nu(1, 5), \nu(1, 3)\} = 22 \text{ MW} \\ \nu^{[2]}(2, 1) &= \min\{\nu(2, 1), \nu(2, 3)\} = 5 \text{ MW} \\ \nu^{[2]}(2, 4) &= \min\{\nu(2, 4), \nu(2, 3)\} = 17 \text{ MW} \\ \nu^{[2]}(2, 5) &= \min\{\nu(2, 5), \nu(2, 3)\} = 17 \text{ MW} \\ \nu^{[2]}(4, 1) &= \min\{\nu(4, 1), \nu(4, 3)\} = 8 \text{ MW} \\ \nu^{[2]}(4, 2) &= \min\{\nu(4, 2), \nu(4, 3)\} = 8 \text{ MW} \\ \nu^{[2]}(4, 5) &= \min\{\nu(4, 5), \nu(4, 3)\} = 5 \text{ MW} \\ \nu^{[2]}(5, 1) &= \min\{\nu(5, 1), \nu(5, 3)\} = 13 \text{ MW} \\ \nu^{[2]}(5, 2) &= \min\{\nu(5, 2), \nu(5, 3)\} = 13 \text{ MW} \\ \nu^{[2]}(5, 4) &= \min\{\nu(5, 4), \nu(5, 3)\} = 5 \text{ MW.} \end{aligned}$$

Thus,

$$\nu^{[2]} = \begin{pmatrix} 0 & 5 & 22 & 22 & 22 \\ 5 & 0 & 17 & 17 & 17 \\ 22 & 17 & 0 & 8 & 13 \\ 8 & 8 & 8 & 0 & 5 \\ 13 & 13 & 13 & 5 & 0 \end{pmatrix} \text{ MW.}$$

In accordance with the updated cost matrix $\nu^{[2]}$, we take from $\Omega_J^{[1]}$ the scenario ω that, if included in the subsequent reduced set $\Omega_S^{[2]}$, minimizes the Kantorovich distance between this set and the original one, Ω . Distances are

$$\begin{aligned} d_1^{[2]} &= \pi_2 \nu^{[2]}(2, 1) + \pi_4 \nu^{[2]}(4, 1) + \pi_5 \nu^{[2]}(5, 1) = 4.30 \text{ MW} \\ d_2^{[2]} &= \pi_1 \nu^{[2]}(1, 2) + \pi_4 \nu^{[2]}(4, 2) + \pi_5 \nu^{[2]}(5, 2) = 4.55 \text{ MW} \\ d_4^{[2]} &= \pi_1 \nu^{[2]}(1, 4) + \pi_2 \nu^{[2]}(2, 4) + \pi_5 \nu^{[2]}(5, 4) = 9.40 \text{ MW} \\ d_5^{[2]} &= \pi_1 \nu^{[2]}(1, 5) + \pi_2 \nu^{[2]}(2, 5) + \pi_4 \nu^{[2]}(4, 5) = 10.15 \text{ MW.} \end{aligned}$$

Hence,

$$\begin{aligned} \Omega_S^{[2]} &= \Omega_S^* = \{1, 3\}, \\ \Omega_J^{[2]} &= \Omega_J^* = \{2, 4, 5\}. \end{aligned}$$

Step 3. Lastly, we add the probabilities of the non-selected scenarios in Ω_J^* to the probabilities of the selected ones in Ω_S^* .

Given that

1. Scenario 1 in Ω_S^* is the *closest* one to scenario 2 in Ω_J^* ($\nu(2, 1) = 5$, while $\nu(2, 3) = 17$),
2. Scenario 3 in Ω_S^* is the *closest* one to scenario 4 in Ω_J^* ($\nu(4, 3) = 8$, while $\nu(4, 1) = 30$), and
3. Scenario 3 in Ω_S^* is the *closest* one to scenario 5 in Ω_J^* ($\nu(5, 3) = 13$, while $\nu(5, 1) = 35$),

it follows that

$$\begin{aligned} \pi_1^* &= \pi_1 + \pi_2 = 0.45, \\ \pi_3^* &= \pi_3 + \pi_4 + \pi_5 = 0.55. \end{aligned}$$

In conclusion, the variant 1 of the forward selection algorithm yields a reduced scenario set $\Omega_S^* = \{1, 3\}$ with associated probabilities $\pi_1^* = 0.45$ and $\pi_3^* = 0.55$.

Solution to Exercise 3.9

We follow the algorithm described step by step in Subsection 3.4.2 of Chapter 3.

Step 1. Fit univariate ARMA models to stochastic processes $\mathbf{Y}^a$ and $\mathbf{Y}^b$.

The fitted models are provided in the statement of the exercise,

$$\begin{pmatrix} y_t^a \\ y_t^b \end{pmatrix} = \begin{pmatrix} 0.908 & 0 \\ 0 & 0.850 \end{pmatrix} \begin{pmatrix} y_{t-1}^a \\ y_{t-1}^b \end{pmatrix} + \begin{pmatrix} \varepsilon_t^a \\ \varepsilon_t^b \end{pmatrix} + \begin{pmatrix} 0.182 & 0 \\ 0 & 0.203 \end{pmatrix} \begin{pmatrix} \varepsilon_{t-1}^a \\ \varepsilon_{t-1}^b \end{pmatrix}.$$

Step 2. Compute variance-covariance matrix $\mathbf{G}$ ($N_T = 2$ and $K = 1$):

$$\mathbf{G}_{11} = 0.50^2 \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.25 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & 0.25 \end{pmatrix}$$

$$\mathbf{G}_{22} = 0.20^2 \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.04 & 0 & 0 \\ 0 & 0.04 & 0 \\ 0 & 0 & 0.04 \end{pmatrix}$$

$$\mathbf{G}_{12} = \mathbf{G}_{21}^T = 0.50 \times 0.20 \times \begin{pmatrix} 0.7 & 0.1 & 0 \\ 0.3 & 0.7 & 0.1 \\ 0 & 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} 0.07 & 0.01 & 0 \\ 0.03 & 0.07 & 0.01 \\ 0 & 0.03 & 0.07 \end{pmatrix}.$$

Therefore,

$$\mathbf{G} = \left(\begin{array}{ccc|ccc} 0.25 & 0 & 0 & 0.07 & 0.01 & 0 \\ 0 & 0.25 & 0 & 0.03 & 0.07 & 0.01 \\ 0 & 0 & 0.25 & 0 & 0.03 & 0.07 \\ \hline 0.07 & 0.03 & 0 & 0.04 & 0 & 0 \\ 0.01 & 0.07 & 0.03 & 0 & 0.04 & 0 \\ 0 & 0.01 & 0.07 & 0 & 0 & 0.04 \end{array} \right).$$

Step 3. Determine orthogonal matrix $\mathbf{L}$ by applying the Cholesky decomposition to matrix $\mathbf{G}$,

$$\mathbf{L} = \left(\begin{array}{ccc|ccc} 0.5000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.5000 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5000 & 0 & 0 & 0 \\ \hline 0.1400 & 0.0600 & 0 & 0.1296 & 0 & 0 \\ 0.0200 & 0.1400 & 0.0600 & -0.0864 & 0.0945 & 0 \\ 0 & 0.0200 & 0.1400 & -0.0093 & -0.1270 & 0.0616 \end{array} \right)$$

Step 4. Simulate $N_T \times N_S = 2 \times 2 = 4$ independent standard normal errors, $\boldsymbol{\xi}$, e.g.,

$$\boldsymbol{\xi} = \begin{pmatrix} \boldsymbol{\xi}^a \\ \boldsymbol{\xi}^b \end{pmatrix} = \begin{pmatrix} \xi_1^a \\ \xi_2^a \\ \xi_1^b \\ \xi_2^b \end{pmatrix} = \begin{pmatrix} -0.4326 \\ -1.6656 \\ 0.1253 \\ 0.2877 \end{pmatrix}.$$

Step 5. Cross-correlate the standard normal errors generated in the previous step by using the orthogonal transformation (3.31) described in Subsection 3.4.2 of Chapter 3,

$$\begin{pmatrix} \varepsilon_0^a \\ \varepsilon_1^a \\ \varepsilon_2^a \\ \varepsilon_0^b \\ \varepsilon_1^b \\ \varepsilon_2^b \end{pmatrix} = \left(\begin{array}{ccc|ccc} 0.5000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.5000 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5000 & 0 & 0 & 0 \\ \hline 0.1400 & 0.0600 & 0 & 0.1296 & 0 & 0 \\ 0.0200 & 0.1400 & 0.0600 & -0.0864 & 0.0945 & 0 \\ 0 & 0.0200 & 0.1400 & -0.0093 & -0.1270 & 0.0616 \end{array} \right) \begin{pmatrix} -0.3000 \\ -0.4326 \\ -1.6656 \\ -0.1200 \\ 0.1253 \\ 0.2877 \end{pmatrix} \\
= \begin{pmatrix} -0.1500 \\ -0.2163 \\ -0.8328 \\ -0.0835 \\ -0.1443 \\ -0.2389 \end{pmatrix}.$$

Step 6. Lastly, introduce the cross-correlated normal errors generated in Step 5 above into the ARMA models fitted in Step 1 so as to simulate two values of stochastic processes $\mathbf{Y}^a$ and $\mathbf{Y}^b$, i.e.,

$$\begin{pmatrix} y_1^a \\ y_1^b \end{pmatrix} = \begin{pmatrix} 0.908 & 0 \\ 0 & 0.850 \end{pmatrix} \begin{pmatrix} y_0^a \\ y_0^b \end{pmatrix} + \begin{pmatrix} \varepsilon_1^a \\ \varepsilon_1^b \end{pmatrix} + \begin{pmatrix} 0.182 & 0 \\ 0 & 0.203 \end{pmatrix} \begin{pmatrix} \varepsilon_0^a \\ \varepsilon_0^b \end{pmatrix} \\
= \begin{pmatrix} 0.908 & 0 \\ 0 & 0.850 \end{pmatrix} \begin{pmatrix} 9.25 \\ 5.60 \end{pmatrix} + \begin{pmatrix} -0.2163 \\ -0.1443 \end{pmatrix} + \begin{pmatrix} 0.182 & 0 \\ 0 & 0.203 \end{pmatrix} \begin{pmatrix} -0.30 \\ -0.12 \end{pmatrix} \\
= \begin{pmatrix} 8.1281 \\ 4.5913 \end{pmatrix}$$

$$\begin{pmatrix} y_2^a \\ y_2^b \end{pmatrix} = \begin{pmatrix} 0.908 & 0 \\ 0 & 0.850 \end{pmatrix} \begin{pmatrix} y_1^a \\ y_1^b \end{pmatrix} + \begin{pmatrix} \varepsilon_2^a \\ \varepsilon_2^b \end{pmatrix} + \begin{pmatrix} 0.182 & 0 \\ 0 & 0.203 \end{pmatrix} \begin{pmatrix} \varepsilon_1^a \\ \varepsilon_1^b \end{pmatrix} \\
= \begin{pmatrix} 0.908 & 0 \\ 0 & 0.850 \end{pmatrix} \begin{pmatrix} 8.1281 \\ 4.5913 \end{pmatrix} + \begin{pmatrix} -0.8328 \\ -0.2389 \end{pmatrix} \\
+ \begin{pmatrix} 0.182 & 0 \\ 0 & 0.203 \end{pmatrix} \begin{pmatrix} -0.2163 \\ -0.1443 \end{pmatrix} = \begin{pmatrix} 6.5081 \\ 3.6344 \end{pmatrix}.$$

In short, the pair of correlated scenarios $(\hat{\mathbf{Y}}^a, \hat{\mathbf{Y}}^b)^T$ spanning two periods are

$$\begin{pmatrix} \hat{\mathbf{Y}}^a \\ \hat{\mathbf{Y}}^b \end{pmatrix} = \begin{pmatrix} 8.1281 & 6.5081 \\ 4.5913 & 3.6344 \end{pmatrix}.$$

C.3 Exercises from Chapter 4

Solution to Exercise 4.1

The formulation of the problem is the following:

$$\begin{aligned} & \text{Maximize}_{P_f^F, v_t, P_{t\omega}^P, P_{t\omega}^G} \\ & \sum_{f=1}^3 \sum_{t=1}^4 \lambda_f^F P_f^F + \sum_{\omega=1}^5 \pi_{\omega} \left[\sum_{t=1}^4 \left(\lambda_{t\omega}^P P_{t\omega}^P - C^G P_{t\omega}^G \right) \right] \end{aligned} \quad (\text{C.3})$$

subject to

$$0 \leq P_f^F \leq 25, \quad f = 1, 2, 3 \quad (\text{C.4})$$

$$\sum_{f=1}^3 P_f^F + P_{t\omega}^P = P_{t\omega}^G, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.5})$$

$$20v_t \leq P_{t\omega}^G \leq 100v_t, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.6})$$

$$P_{t\omega}^P \geq 0, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.7})$$

$$v_t \in \{0, 1\}, \quad t = 1, \dots, 4, \quad (\text{C.8})$$

where P_f^F is the power purchased from forward contract f , $P_{t\omega}^P$ and $P_{t\omega}^G$ are the power sold in the pool and the power generated in period t and scenario ω , respectively, v_t is a binary variable that is equal to 1 if the unit is online in period t , being 0 otherwise, λ_f^F is the selling price of contract f , $\lambda_{t\omega}^P$ is the pool price in period t and scenario ω , C^G is the linear production cost of the unit, and π_{ω} is the probability of scenario ω .

The optimal solution to this problem is obtained for $\{P_1^{F*}, P_2^{F*}, P_3^{F*}\} = \{0, 0, 0\}$ and $\{v_1^*, v_2^*, v_3^*, v_4^*\} = \{0, 1, 1, 1\}$. The expected profit is equal to \$734.4.

The profit per scenario is provided in Table C.2.

Table C.2 Solution to Exercise 4.1: profit per scenario

Scenario #	Profit (\$)
1	534.0
2	1280.0
3	258.0
4	420.0
5	1180.0

Solution to Exercise 4.5

The formulation of the problem described in Exercise 4.1 including the VaR is the following:

$$\begin{aligned} & \text{Maximize}_{P_f^F, v_t, P_{t\omega}^P, P_{t\omega}^G, \eta, \theta_\omega} \\ (1 - \beta) & \left[\sum_{f=1}^3 \sum_{t=1}^4 \lambda_f^F P_f^F + \sum_{\omega=1}^5 \pi_\omega \sum_{t=1}^4 \left(\lambda_{t\omega}^P P_{t\omega}^P - C^G P_{t\omega}^G \right) \right] + \beta \eta \end{aligned} \quad (\text{C.9})$$

subject to

$$0 \leq P_f^F \leq 25, \quad f = 1, 2, 3 \quad (\text{C.10})$$

$$\sum_{f=1}^3 P_f^F + P_{t\omega}^P = P_{t\omega}^G, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.11})$$

$$20v_t \leq P_{t\omega}^G \leq 100v_t, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.12})$$

$$\sum_{\omega=1}^5 \pi_\omega \theta_\omega \leq 1 - \alpha \quad (\text{C.13})$$

$$\eta - \left[\sum_{f=1}^3 \sum_{t=1}^4 \lambda_f^F P_f^F + \sum_{t=1}^4 \left(\lambda_{t\omega}^P P_{t\omega}^P - C^G P_{t\omega}^G \right) \right] \leq M\theta_\omega, \quad \omega = 1, \dots, 5 \quad (\text{C.14})$$

$$P_{t\omega}^P \geq 0, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.15})$$

$$v_t \in \{0, 1\}, \quad t = 1, \dots, 4 \quad (\text{C.16})$$

$$\theta_\omega \in \{0, 1\}, \quad \omega = 1, \dots, 5. \quad (\text{C.17})$$

The results in terms of the expected profit, the VaR, and the first-stage variables for different values of β are provided in Table C.3. Observe that the expected profit decreases as the VaR increases and that the risk is reduced by using the forward contracts. Note that selling energy through a forward contract requires that the unit of the producer must be online in all periods within the time span of the contract (4 periods).

Table C.3 Exercise 4.3: results

β	Expected profit (\$)	VaR (\$)	P_1^F (MW)	P_2^F (MW)	P_3^F (MW)	v_1	v_2	v_3	v_4
0	734.4	420.0	0	0	0	0	1	1	1
0.12	726.4	488.0	0	0	20	1	1	1	1
1	718.5	495.0	0	0	25	1	1	1	1

Solution to Exercise 4.8

The formulation of the problem described in Exercise 4.1 including second-order stochastic dominance constraints is

$$\begin{aligned} & \text{Maximize}_{P_f^F, v_t, P_{t\omega}^P, P_{t\omega}^G, s_{\omega v}} \\ & \sum_{f=1}^3 \sum_{t=1}^4 \lambda_f^F P_f^F + \sum_{\omega=1}^5 \pi_{\omega} \sum_{t=1}^4 \left(\lambda_{t\omega}^P P_{t\omega}^P - C^G P_{t\omega}^G \right) \end{aligned} \quad (\text{C.18})$$

subject to

$$0 \leq P_f^F \leq 25, \quad f = 1, 2, 3 \quad (\text{C.19})$$

$$\sum_{f=1}^3 P_f^F + P_{t\omega}^P = P_{t\omega}^G, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.20})$$

$$20v_t \leq P_{t\omega}^G \leq 100v_t, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.21})$$

$$\begin{aligned} \eta - \left[\sum_{f=1}^3 \sum_{t=1}^4 \lambda_f^F P_f^F + \sum_{t=1}^4 \left(\lambda_{t\omega}^P P_{t\omega}^P - C^G P_{t\omega}^G \right) \right] & \leq s_{\omega v}, \\ \omega = 1, \dots, 5; v = 1, \dots, 4 \end{aligned} \quad (\text{C.22})$$

$$\sum_{\omega=1}^5 \pi_{\omega} s_{\omega v} \leq \sum_{v'=1}^4 \tau'_{v'} \max(k_v - k_{v'}, 0), \quad v = 1, \dots, 5 \quad (\text{C.23})$$

$$P_{t\omega}^P \geq 0, \quad t = 1, \dots, 4; \omega = 1, \dots, 5 \quad (\text{C.24})$$

$$v_t \in \{0, 1\}, \quad t = 1, \dots, 4 \quad (\text{C.25})$$

$$s_{\omega, v} \geq 0, \quad \omega = 1, \dots, 5; v = 1, \dots, 4. \quad (\text{C.26})$$

In this exercise two benchmarks with four scenarios are used. The probability and the value of the benchmark for each scenario, $\{\tau_v, k_v\}$, $v = 1, \dots, 4$, are provided in Table C.4. Observe that the second benchmark is identical to the first one but displaced \$50 *to the right*. Therefore, we can say that benchmark 2 is more restrictive than benchmark 1.

Table C.4 Solution to Exercise 4.8: benchmark data

Scenario #	Probability	Benchmark 1	Benchmark 2
		(\$)	(\$)
1	0.25	300	350
2	0.25	450	500
3	0.25	600	650
4	0.25	700	750

Imposing each benchmark, the resulting profits per scenario are provided in Table C.5. The expected profit and the optimal first-stage variables are provided in Table C.6. Note that imposing benchmark 2 yields a smaller expected profit. This result is coherent since benchmark 2 is more restrictive

than benchmark 1. Observe also that the profit associated with the worst scenarios (scenarios $\{1,3\}$) is higher in the solution to benchmark 2, at the cost of a smaller profit in the best scenarios (scenarios $\{2,4,5\}$).

Table C.5 Solution to Exercise 4.8: profit per scenario for benchmarks 1 and 2

Scenario #	Profit (\$)	
	Benchmark 1	Benchmark 2
1	560.00	562.12
2	1144.00	1115.15
3	336.00	350.00
4	488.00	493.94
5	1104.00	1077.27

Table C.6 Solution to Exercise 4.8: expected profit and first-stage decisions for benchmarks 1 and 2

Benchmark #	Expected profit (\$)	P_1^F	P_2^F	P_3^F	v_1	v_2	v_3	v_4
		(MW)	(MW)	(MW)				
1	726.4	0	0	20.0	1	1	1	1
2	719.7	0	0	24.2	1	1	1	1

C.4 Exercises from Chapter 5

Solution to Exercise 5.1

First, observe that considering the producer as a price-taker agent in the adjustment market implies that $\lambda_{t\omega}^A = \lambda_{t\omega}^{A0}$ and $\gamma_{t\omega}^A = 0$. In addition, this assumption causes that variable $P_{t\omega}^A$ is no longer needed. As a result, the objective function and the set of constraints modeling the CVaR in problem (5.48)-(5.63) must be modified. This way, the objective function (5.48) results in

$$\begin{aligned}
& \text{Maximize}_{P_{gt\omega}^A, P_{gt\omega}^D, P_{gt\omega}^G, P_{gt\omega}^R, w_{gt\omega}, \zeta, \eta_\omega} \\
& \sum_{\omega=1}^{N_\Omega} \pi_\omega \sum_{t=1}^{N_T} \left[\sum_{g=1}^{N_G} \lambda_{t\omega}^D P_{gt\omega}^D + \sum_{g=1}^{N_G} \lambda_{t\omega}^R P_{gt\omega}^R + \sum_{g=1}^{N_G} \lambda_{t\omega}^A P_{gt\omega}^A \right. \\
& \left. - \sum_{g=1}^{N_G} C_g^P (P_{gt\omega}^D + P_{gt\omega}^R) \right] + \beta \left(\zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{N_\Omega} \pi_\omega \eta_\omega \right), \quad (C.27)
\end{aligned}$$

whereas the CVaR constraints (5.60) yield

$$\begin{aligned}
& \zeta - \sum_{t=1}^{N_T} \left[\sum_{g=1}^{N_G} \lambda_{t\omega}^D P_{gt\omega}^D + \sum_{g=1}^{N_G} \lambda_{t\omega}^R P_{gt\omega}^R + \sum_{g=1}^{N_G} \lambda_{t\omega}^A P_{gt\omega}^A - \right. \\
& \left. \sum_{g=1}^{N_G} C_g^P (P_{gt\omega}^D + P_{gt\omega}^A) \right] \leq \eta_\omega, \quad \forall \omega. \quad (C.28)
\end{aligned}$$

Observe also that constraint (5.54), which computes the value of variable $P_{t\omega}^A$, is not needed and can be removed from the final formulation of the problem.

In order to extend formulation (5.48)-(5.63) for considering offering curves instead of single power quantities in both the regulation and the adjustment markets, it is necessary to relax the non-anticipativity constraints (5.52) and (5.53). Thus, we replace vectors $\mathbf{A}^R$ and $\mathbf{A}^A$ in (5.52) and (5.53) by the new vectors $\mathbf{A}^{R'}$ and $\mathbf{A}^{A'}$, respectively. This way, $\mathbf{A}^{R'}$ is a vector whose element $A^{R'}(\omega)$ is equal to 1 if scenarios ω and $\omega + 1$ have identical regulation prices. In the same way $\mathbf{A}^{A'}$ is a vector whose element $A^{A'}(\omega)$ is equal to 1 if scenarios ω and $\omega + 1$ have identical adjustment prices. Then, the resulting non-anticipativity constraints are

$$P_{gt\omega}^R - P_{gt\omega+1}^R = 0, \quad \forall g, \forall t, \forall \omega; \text{ if } A^{R'}(\omega) = 1 \quad (C.29)$$

$$P_{gt\omega}^A - P_{gt\omega+1}^A = 0, \quad \forall g, \forall t, \forall \omega; \text{ if } A^{A'}(\omega) = 1. \quad (C.30)$$

Additionally, it is necessary to enforce that the resulting offering curves in the regulation and the adjustment markets must be non-decreasing. This can be stated using an analogous formulation to that used for modeling the offering curves in the day-ahead market (Subsection 5.4.1 of Chapter 5). The resulting formulation for the non-decreasing offering curves in the regulation and the adjustment markets is

$$P_{gt\omega}^R - P_{gt\omega'}^R \leq 0, \quad \forall g, \forall t, \forall \omega; \text{ if } O^R(t, \omega) + 1 = O^R(t, \omega'), \text{ if } Q^D(\omega, \omega') = 1 \quad (C.31)$$

$$P_{gt\omega}^A - P_{gt\omega'}^A \leq 0, \quad \forall g, \forall t, \forall \omega; \text{ if } O^A(t, \omega) + 1 = O^A(t, \omega'), \text{ if } Q^R(\omega, \omega') = 1, \quad (C.32)$$

where $\mathbf{O}^R$ and $\mathbf{O}^A$ are matrices containing the ranking for the regulation and the adjustment prices, respectively, (analogous to matrix $\mathbf{O}^D$ defined for the day-ahead price ranking in Subsection 5.4.1 of Chapter 5), $\mathbf{Q}^D$ is a matrix whose element $Q^D(\omega, \omega')$ is equal to 1 if the day-ahead prices in scenarios ω and ω' are identical; and, in the same manner, $\mathbf{Q}^R$ is a matrix whose element $Q^R(\omega, \omega')$ is equal to 1 if the regulation prices in scenarios ω and ω' are the same. This formulation allows obtaining different offering curves in the regulation market for different realizations of the day-ahead prices. Likewise, for each realization of the regulation prices a different offering curve in the adjustment market is derived.

Solution to Exercise 5.2

The on/off status of each producer unit is modeled by means of the binary variable v_{gt} , which is equal to 1 if unit g is online in period t , being 0 otherwise. Here, we define this variable as a here-and-now decision, being its value equal for all possible scenarios.

In order to account for the binary variable v_{gt} , constraints (5.55) and (5.56) should be replaced by the following ones:

$$P_{gt\omega}^D + \frac{1}{2}P_{gt\omega}^R + P_{gt\omega}^A \leq P_g^{G,\max}v_{gt}, \quad \forall g, \forall t, \forall \omega \quad (\text{C.33})$$

$$P_{gt\omega}^D - \frac{1}{2}P_{gt\omega}^R + P_{gt\omega}^A \geq P_g^{G,\min}v_{gt}, \quad \forall g, \forall t, \forall \omega. \quad (\text{C.34})$$

If the on/off status of the units is accounted for, the start-up and shutdown costs should be also considered. The start-up cost is formulated as

$$C_{gt}^U \geq C_g^{U0}(v_{gt} - v_{gt-1}), \quad \forall g, \forall t \quad (\text{C.35})$$

$$C_{gt}^U \geq 0, \quad \forall g, \forall t, \quad (\text{C.36})$$

where C_{gt}^U is a variable modeling the start-up cost incurred by unit g in period t , and C_g^{U0} is a parameter equal to the start-up cost of unit g .

The shutdown cost is

$$C_{gt}^D \geq C_g^{D0}(v_{gt-1} - v_{gt}), \quad \forall g, \forall t \quad (\text{C.37})$$

$$C_{gt}^D \geq 0, \quad \forall g, \forall t, \quad (\text{C.38})$$

where C_{gt}^D is a variable modeling the shutdown cost incurred by unit g in period t , and C_g^{D0} is a parameter equal to the shutdown cost of unit g .

The start-up and shutdown costs have to be included in the objective function and the CVaR constraints of problem (5.48)-(5.63). The objective function considering start-up and shutdown costs is

$$\begin{aligned}
& \text{Maximize}_{C_{gt}^D, C_{gt}^U, P_{t\omega}^A, P_{gt\omega}^A, P_{gt\omega}^D, P_{gt\omega}^G, P_{gt\omega}^R, u_{gt}, w_{gt\omega}, \zeta, \eta_\omega} \\
& \sum_{\omega=1}^{N_\Omega} \pi_\omega \sum_{t=1}^{N_T} \left[\sum_{g=1}^{N_G} \lambda_{t\omega}^D P_{gt\omega}^D + \sum_{g=1}^{N_G} \lambda_{t\omega}^R P_{gt\omega}^R + \lambda_{t\omega}^{A0} P_{t\omega}^A + \gamma_{t\omega}^A (P_{t\omega}^A)^2 \right. \\
& \left. - \sum_{g=1}^{N_G} \left(C_g^P (P_{gt\omega}^D + P_{gt\omega}^R) + C_{gt}^U + C_{gt}^D \right) \right] + \beta \left(\zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{N_\Omega} \pi_\omega \eta_\omega \right).
\end{aligned} \tag{C.39}$$

The resulting CVaR constraints are

$$\begin{aligned}
& \zeta - \sum_{t=1}^{N_T} \left[\sum_{g=1}^{N_G} \lambda_{t\omega}^D P_{gt\omega}^D + \sum_{g=1}^{N_G} \lambda_{t\omega}^R P_{gt\omega}^R + \lambda_{t\omega}^{A0} P_{t\omega}^A + \gamma_{t\omega}^A (P_{t\omega}^A)^2 - \right. \\
& \left. \sum_{g=1}^{N_G} \left(C_g^P (P_{gt\omega}^D + P_{gt\omega}^R) + C_{gt}^U + C_{gt}^D \right) \right] \leq \eta_\omega, \forall \omega.
\end{aligned} \tag{C.40}$$

Solution to Exercise 5.3

The second adjustment market is an additional trading floor in which producers can submit offering curves after the first adjustment market is cleared. Traditionally, the volume of energy associated to the successive adjustment markets is smaller as the time between the market clearing and the energy delivery decreases. The decision-making framework for the producer pool problem including a second adjustment market is provided in Fig. C.4. Observe that different adjustment markets may have associated different planning horizon. For instance, in this exercise the first adjustment market comprises all hours of the next day (24), whereas the second one only covers the last 12 hours.

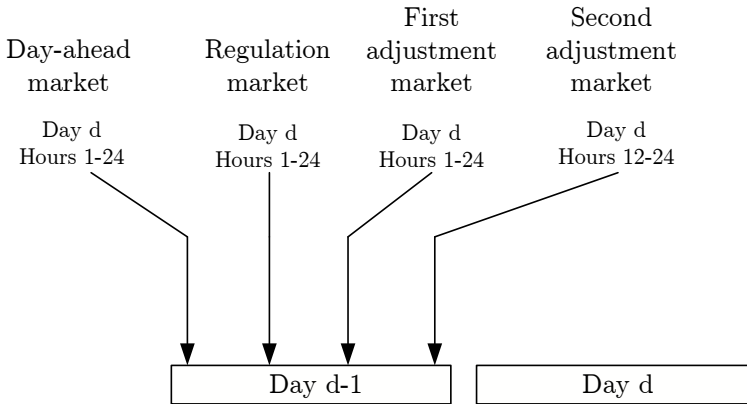


Fig. C.4 Solution to Exercise 5.3: decision framework

If we include a second adjustment market, the decision-making process changes with respect to the case with a single adjustment market. In fact, an additional stage has to be included in the stochastic programming problem in order to account for the decisions associated with the second adjustment market. Fig. C.5 provides the scenario tree for the decision-making process including day-ahead, regulation, first adjustment, and second adjustment markets.

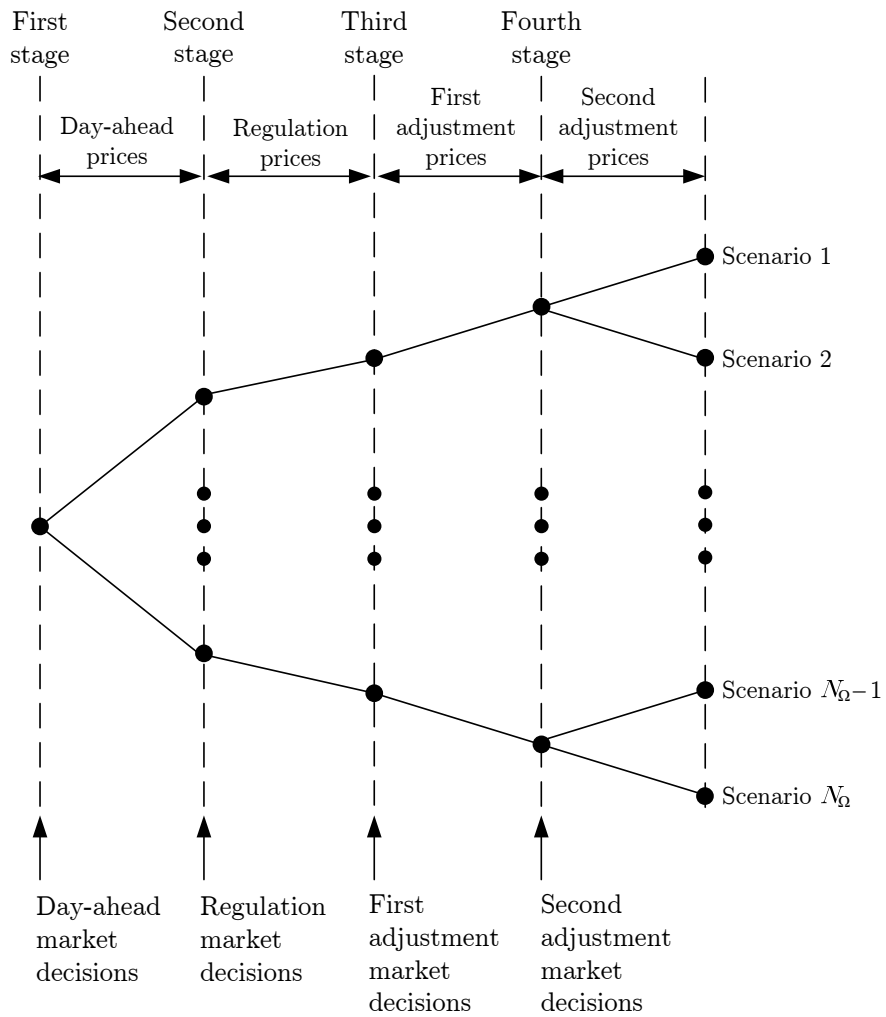


Fig. C.5 Solution to Exercise 5.3: scenario tree

C.5 Exercises from Chapter 6

Solution to Exercise 6.3

According to equation (6.15) in Subsection 6.3.2 of Chapter 6, the imbalance cost C_t incurred by the wind power producer in period t is given by

$$C_t = \lambda_t^D(1 - r_t^+) \Delta_t = \$24/\text{MWh} \times (1 - 0.85) \times (100 - 75)\text{MWh} = \$90,$$

since $\Delta_t = 100 - 75 = 25 \text{ MWh} > 0$. As a result, the profit obtained from the energy trading in this period is

$$R_t = \lambda_t^D E_t - C_t = \$24/\text{MWh} \times 100 \text{ MWh} - \$90 = \$2310.$$

Solution to Exercise 6.5

The scenario tree built from the statistical information on the day-ahead market price, the price difference between day-ahead and adjustment markets, the imbalance price ratios and the wind power production provided in the statement of the exercise is depicted in Fig. C.6. Prices are in \$/MWh, powers in MW and ratios are non-dimensional numbers. For illustrative purposes, we detail below the computation of some of the probabilities π_ω :

$$\begin{aligned} \pi_1 &= 0.30 \times 0.60 \times 0.50 \times 0.50 \times 0.45 = 0.02025 \\ \pi_2 &= 0.30 \times 0.60 \times 0.50 \times 0.50 \times 0.55 = 0.02475 \\ &\vdots \\ \pi_9 &= 0.30 \times 0.40 \times 0.50 \times 0.35 \times 0.45 = 0.00945 \\ \pi_{10} &= 0.30 \times 0.40 \times 0.50 \times 0.35 \times 0.55 = 0.01155 \\ \pi_{11} &= 0.30 \times 0.40 \times 0.50 \times 0.65 \times 0.45 = 0.01755 \\ \pi_{12} &= 0.30 \times 0.40 \times 0.50 \times 0.65 \times 0.55 = 0.02145 \\ &\vdots \\ \pi_{17} &= 0.70 \times 0.60 \times 0.50 \times 0.50 \times 0.45 = 0.04725 \\ \pi_{18} &= 0.70 \times 0.60 \times 0.50 \times 0.50 \times 0.55 = 0.05775 \\ &\vdots \\ \pi_{29} &= 0.70 \times 0.40 \times 0.50 \times 0.35 \times 0.45 = 0.02205 \\ \pi_{30} &= 0.70 \times 0.40 \times 0.50 \times 0.35 \times 0.55 = 0.02695 \\ \pi_{31} &= 0.70 \times 0.40 \times 0.50 \times 0.65 \times 0.45 = 0.04095 \\ \pi_{32} &= 0.70 \times 0.40 \times 0.50 \times 0.65 \times 0.55 = 0.05005. \end{aligned}$$

Solution to Exercise 6.7

The offering strategy model (6.36)–(6.41) in Chapter 6 does not take into account the existence of an adjustment market and as a result, the 32 scenarios in the tree represented in Fig. C.6 can be reduced to 16 as shown in Fig. C.7.

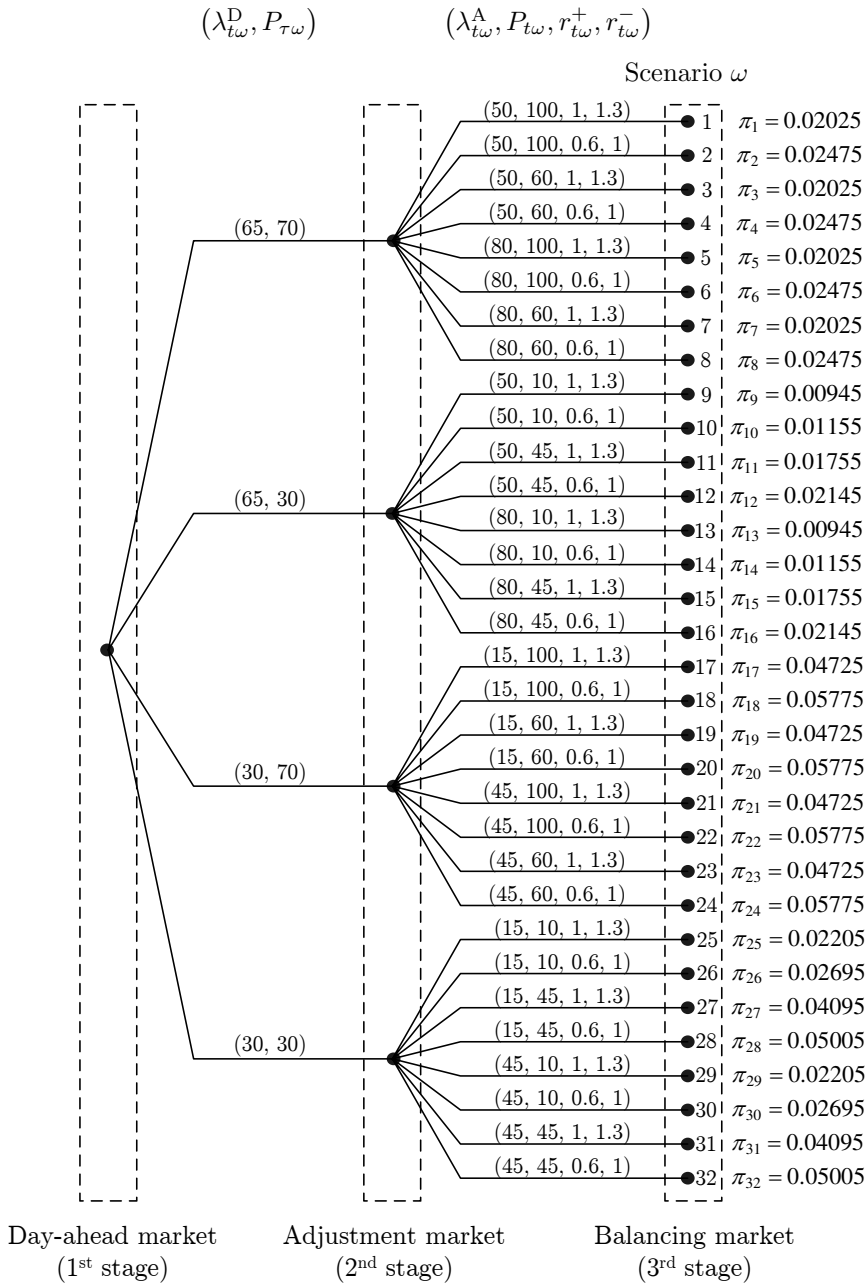


Fig. C.6 Solution to Exercise 6.5: scenario tree

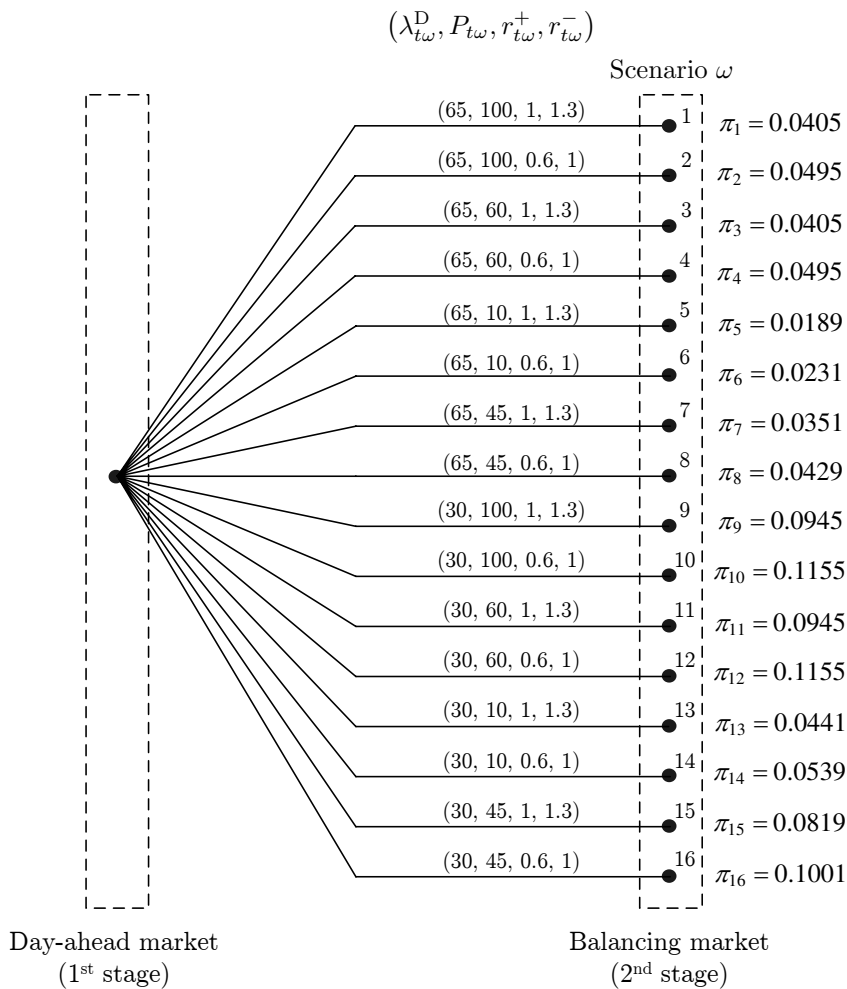


Fig. C.7 Solution to Exercise 6.7: two-stage scenario tree (excluding the adjustment market)

The deterministic approach to solve the wind producer problem simply comes down to selling the expected wind power production ($\hat{P}$) in the day-ahead market, which is computed as follows:

$$\begin{aligned}
\hat{P} &= \xi\{P\} = \sum_{\omega=1}^{16} \pi_{\omega} P_{\omega} = (0.0405 + 0.0495 + 0.0945 + 0.1155) \times 100 \\
&+ (0.0405 + 0.0495 + 0.0945 + 0.1155) \times 60 \\
&+ (0.0189 + 0.0231 + 0.0441 + 0.0539) \times 10 \\
&+ (0.0351 + 0.0429 + 0.0819 + 0.1001) \times 45 = 61.10 \text{ MW}.
\end{aligned}$$

If we fix decision variable P^D in model (6.36)–(6.41) to 61.10 MW, the resulting expected profit (z^{D*}) is \$2306.8. Therefore, the Value of Stochastic Solution is

$$\text{VSS} = z^{S*} - z^{D*} = 2308 - 2306.8 = \$1.2$$

where z^{S*} is the expected profit associated with the stochastic solution calculated in Exercise 6.6. The VSS represents a percentage of 0.052% with respect to z^{S*} .

Finally, Table C.7 provides the expected values of perfect information as improvements in percentage with respect to the expected profit resulting from model (6.36)–(6.41).

Table C.7 Solution to Exercise 6.7: expected values of perfect information

Perfect information on	λ^D	P	(r^+, r^-)
EVPI (%)	0	7.22	7.22

C.6 Exercises from Chapter 7

Solution to Exercise 7.1

Constraints (7.11) of Chapter 7 are reformulated as

$$u_{gt\omega} P_g^{G,\min} a_{gt\omega} d_t \leq E_{gt\omega}^G \leq u_{gt\omega} P_g^{G,\max} a_{gt\omega} d_t, \forall g, \forall t, \forall \omega, \quad (\text{C.41})$$

where $u_{gt\omega}$ is a binary (0/1) variable modeling the off/on status of unit g during period t and scenario ω , and $P_g^{G,\min}$ is the minimum power output of unit g .

It should be noted that expressions (C.41) are linear but involve binary variables.

Additionally, note that if the start-up costs of the units are incorporated in the objective function of the forward contracting problem of a producer, problem (7.18)–(7.25), constraints (C.41) may render obsolete constraints (7.21).

Solution to Exercise 7.4

The formulation of the forward contracting problem for a producer avoiding parameter β is provided below,

$$\begin{aligned} & \text{Maximize}_{P_{fj}^F, E_{t\omega}^P, E_{gt\omega}^G, \zeta, \eta_\omega} \\ & \zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{N_\Omega} \pi_\omega \eta_\omega \end{aligned} \quad (\text{C.42})$$

subject to

$$0 \leq P_{fj}^F \leq \bar{P}_{fj}^F, \quad \forall f, \forall j \quad (\text{C.43})$$

$$0 \leq E_{gt\omega}^G \leq P_g^{G, \max} d_t, \quad \forall g, \forall t, \forall \omega \quad (\text{C.44})$$

$$E_{gt\omega}^G \geq H_g E_{gt'\omega}^G, \quad \forall g, \forall \omega, \forall t, t' \mid W(t, t') = 1 \quad (\text{C.45})$$

$$\sum_{g=1}^{N_G} E_{gt\omega}^G = E_{t\omega}^P + \sum_{f \in F_t} \sum_{j=1}^{N_J} P_{fj}^F d_t + E_t^{\text{PC}}, \quad \forall t, \forall \omega \quad (\text{C.46})$$

$$\zeta - \sum_{t=1}^{N_T} \left(\sum_{f \in F_t} \sum_{j=1}^{N_J} \lambda_{fj}^F P_{fj}^F d_t + \lambda_{t\omega}^P E_{t\omega}^P - \sum_{g=1}^{N_G} C_g^G E_{gt\omega}^G \right) \leq \eta_\omega, \quad \forall \omega \quad (\text{C.47})$$

$$\eta_\omega \geq 0, \quad \forall \omega \quad (\text{C.48})$$

$$E_{t\omega}^P \geq 0, \quad \forall t, \forall \omega. \quad (\text{C.49})$$

Observe that the objective function of the problem above is the CVaR of the profit distribution. The interested reader should check the definition of the CVaR in Chapter 3.

To manage risk using the above formulation, the parameter α is used. A value of $\alpha = 0$ provides the risk-neutral decision (i.e., the expected value of the profit is maximized), while an α value close to 1 involves a risk-averse decision, which avoids scenarios of low profit at the cost of reducing the expected profit.

Solution to Exercise 7.7

Table C.8 provides the results for different values of β and all possible pairs of scenarios. In Table 7.8 of Section 7.7 of Chapter 7, it can be observed that the average values of the four pool price scenarios are $\{35.25, 32.54, 36.75, 35.75\}$, measured in \$/MWh. Therefore, scenarios $\{3, 4\}$ contain the highest pool prices, whereas the lowest ones are associated with scenarios $\{1, 2\}$. For this reason, in the case considering scenarios $\{3, 4\}$ the forward contracts are not signed (forward contracts are not competitive with respect to the pool). The same fact occurs for pairs of scenarios $\{1, 3\}$ and $\{1, 4\}$. On the contrary, in cases with scenarios $\{1, 2\}$, $\{2, 3\}$, and $\{2, 4\}$, the producer sells power through forward contracts for all values of β .

Solution to Exercise 7.8

Table C.8 Solution to Exercise 7.7: expected profit and CVaR for different values of β and pairs of scenarios

Scenarios	β	Expected profit (\$)	CVaR (\$)	Power sold through contract 1 (MW)	Power sold through contract 2 (MW)
1-2	0	7436.00	5962.00	100	82
	10	7117.50	6117.50	200	100
1-3	0	11,088.75	9254.50	0	0
	10	11,088.75	9254.50	0	0
1-4	0	10,357.25	9254.50	0	0
	10	10,357.25	9254.50	0	0
2-3	0	8887.52	5552.03	100	0
	10	7992.50	6117.50	200	100
2-4	0	8357.25	5962.00	100	82
	10	7742.50	6117.50	200	100
3-4	0	12,191.50	11,460.00	0	0
	10	12,191.50	11,460.00	0	0

Table C.9 provides the results for different values of β and for three cases considering (i) all contracts, (ii) only contract 1, and (iii) only contract 2. Observe that for $\beta = 0$ the contracts are not used in any case. For this reason, the results in terms of the expected profit and the CVaR are the same in the three cases. For the rest of cases, observe that the best solution (highest expected profit and highest CVaR) is always obtained if the two contracts are considered. Notice also that the results obtained only considering the first contract outperforms those obtained with only the second contract. This result is a consequence of the higher selling price of the first contract (see Table 7.7 in Section 7.7 of Chapter 7).

Solution to Exercise 7.9

First of all, note that computing EVPI or VSS requires a risk-neutral formulation, which leads to $\beta = 0$.

Considering $\beta = 0$ and solving the particular instance of problem (7.18)-(7.25) that corresponds to the example in Section 7.7 of Chapter 7 renders an objective function optimal value $\Pi^{S*} = \$9548.13$

To compute EVPI, non-anticipativity constraints need to be relaxed in the problem above. Doing so results in an objective function optimal value $\Pi^{P*} = \$9938.75$.

EVPI is then computed as $\Pi^{P*} - \Pi^{S*} = \$390.62$.

To compute VSS two problems need to be solved. The first one involves solving the considered instance of problem (7.18)-(7.25) for the average scenario (just that scenario). The solution to this deterministic problem allows deriving the optimal value of the first-stage variables, i.e., the forward con-

Table C.9 Solution to Exercise 7.8: expected profit and CVaR for different values of β and numbers of available contracts

Available contracts	β	Expected profit (\$)	CVaR (\$)	Power sold through contract 1 (MW)	Power sold through contract 2 (MW)
1-2	0	9548.13	4555.00	0	0
	1	9278.50	5962.00	100	82
	10	8367.50	6117.50	200	100
1	0	9548.13	4555.00	0	0
	1	9522.38	5552.03	100	0
	10	8967.50	5817.50	200	0
2	0	9548.13	4555.00	0	0
	1	9322.38	5352.03	0	100
	10	9322.38	5352.03	0	100

tracting decisions. In this case, the power sold through forward contracts is equal to 0.

Then, the considered instance of problem (7.18)-(7.25) is solved fixing the first-stage variables to the optimal values obtained from solving the deterministic problem above. The objective function optimal value of this problem is $\Pi^{D*} = \$9548.13$, which is the same value obtained from solving the stochastic problem (7.18)-(7.25) for $\beta = 0$. Therefore, the VSS is computed as $\Pi^{S*} - \Pi^{D*} = 0$. This result is a consequence of the fact that the optimal quantities of power sold through forward contracts resulting from solving both the deterministic and the risk-neutral stochastic problems are equal to zero. This result is coherent since the forward contracts are tools used by the market agents in order to reduce the risk of profit variability, which is not accounted for in neither the deterministic nor the risk-neutral stochastic problems.

C.7 Exercises from Chapter 8

Solution to Exercise 8.3

Problem (8.18)-(8.25) can be equivalently formulated by replacing the objective function (8.18) by a new objective function including the CVaR of the profit. The resulting formulation is

$$\begin{aligned} & \text{Maximize}_{P_{fj}^F, \lambda_{ei}^R, v_{ei}, E_{t\omega}^P, \zeta, \eta_\omega} \\ & \zeta - \frac{1}{1-\alpha} \sum_{\omega=1}^{N_\Omega} \pi_\omega \eta_\omega \end{aligned} \quad (\text{C.50})$$

subject to

$$0 \leq P_{fj}^F \leq \bar{P}_{fj}^F, \quad \forall f, \forall j \quad (\text{C.51})$$

$$\bar{\lambda}_{ei-1}^R v_{ei} \leq \lambda_{ei}^R \leq \bar{\lambda}_{ei}^R v_{ei}, \quad \forall e, \forall i \quad (\text{C.52})$$

$$\sum_{i=1}^{N_I} v_{ei} = 1, \quad \forall e \quad (\text{C.53})$$

$$\sum_{e=1}^{N_E} \sum_{i=1}^{N_I} \bar{E}_{eti\omega}^R v_{ei} = E_{t\omega}^P + \sum_{f \in F_t} P_f^F d_t + E_t^{\text{PC}}, \quad \forall t, \forall \omega, \quad (\text{C.54})$$

$$\zeta - \sum_{t=1}^{N_T} \left(\sum_{e=1}^{N_E} \sum_{i=1}^{N_I} \lambda_{ei}^R \bar{E}_{eti\omega}^R - \lambda_{t\omega}^P E_{t\omega}^P - \sum_{f \in F_t} \sum_{j=1}^{N_J} \lambda_{fj}^F P_{fj}^F d_t \right) \leq \eta_\omega, \quad \forall \omega \quad (\text{C.55})$$

$$v_{ei} \in \{0, 1\}, \quad \forall e, \forall i \quad (\text{C.56})$$

$$\eta_\omega \geq 0, \quad \forall \omega. \quad (\text{C.57})$$

Observe that the objective function (8.18) presented in Section 8.6 of Chapter 8 comprises the expected profit plus the CVaR multiplied by β . However, in (C.50), there is a single term modeling the CVaR. In this situation the tradeoff between expected profit and CVaR is not materialized by means of the parameter β , but by varying the confidence level α . This way, if $\alpha = 0$ the objective function is equivalent to maximizing the expected profit. This results from the definition of the CVaR, which can be approximately defined as the expected profit for the $(1-\alpha) \times 100\%$ “worst” scenarios. Therefore, if $\alpha = 0$, the $(1-\alpha) \times 100\% = 100\%$ “worst” scenarios correspond to the whole set of scenarios. The value $\alpha = 0$ corresponds to the case $\beta = 0$ in problem (8.18)-(8.25). On the other hand, for values of α higher than 0, the $\alpha \times 100\%$ “best” profit scenarios of the profit distribution are disregarded from the objective function, and the optimal decision vector is only determined considering the “worst” zone of the profit distribution. In summary, values of α close to 0 yield solutions involving low risk aversion, whereas values of α close to 1 result in risk-averse solutions.

Solution to Exercise 8.8

If the price-quota curve in Fig. 8.11 of Chapter 8 is *shifted to the right* \$20/MWh, the retailer supplies the 100% of the client demand for selling prices up to \$85/MWh. For prices between \$85/MWh and \$95/MWh, it supplies the 35% of the demand, and the 15% for prices between \$95/MWh and

\$105/MWh. In this new case, the solutions attained in terms of the expected profit and the CVaR for different values of β are provided in Table C.10. Observe that the expected profit and the CVaR are significantly higher than those reported in Table 8.5 of Section 8.7 of Chapter 8. This is a consequence of the fact that the clients are willing to pay an additional amount of \$20/MWh with respect to the original case. Since the pool and the forward prices remain equal to those in the original case, the profit obtained by the retailer is significantly higher.

Table C.10 Solution to Exercise 8.8: expected profit and CVaR for different values of β

β	Expected profit (\$)	CVaR (\$)
0	15,605.0	9990.0
1	15,458.3	10,190.0
2	15,271.7	10,350.0
5	14,993.0	10,418.0
15	14,672.3	10,444.0

The power purchased through forward contracts and the selling price offered to the clients is provided in Table C.11. Observe that the reduction in the risk faced by the retailer is achieved by increasing only the forward contract purchases whereas the selling price remains stable. This is due to the high price that the clients are willing to pay, which is higher than the purchase prices in the pool and the futures market.

Table C.11 Solution to Exercise 8.8: forward contract purchases and selling price

β	Contract 1 (MW)	Contract 2 (MW)	Selling price (\$/MWh)
0	0	0	85
1	20	0	85
2	20	20	85
5	40	20	85
15	40	40	85

Solution to Exercise 8.9

In this exercise we consider an additional group of clients whose demand scenarios are provided in Table C.12. The price-quota curve of this group of clients includes 3 blocks. In the first block the 100% of the demand is supplied and it comprises selling prices smaller than \$60/MWh. In the second block the 75% of the demand is provided and the selling prices lie between \$60/MWh

and \$70/MWh. Finally, the third block contains the selling prices between \$70/MWh and \$80/MWh, and it has associated the 25% of the demand.

Table C.12 Solution to Exercise 8.9: demand scenarios for the second client group (MWh)

Scenario #	Period	
	1	2
1	225	175
2	240	215
3	200	235
4	240	215

Considering the formulation of problem (8.18)-(8.25) and the data characterizing the second group of clients, the results in terms of the expected profit and the CVaR for different values of β are provided in Table C.13. Observe that considering an additional group of clients causes the expected profit and the CVaR to increase with respect to their values in the example of Section 8.7 of Chapter 8. In Table C.14, the forward contract purchases and the selling prices for the two client groups are provided.

Table C.13 Solution to Exercise 8.9: expected profit and CVaR for different values of β

β	Expected profit (\$)	CVaR (\$)
0	5391.3	729.0
1	4737.2	2181.5
2	3873.7	2863.5
5	3595.0	2931.5
15	3274.3	2957.5

C.8 Exercises from Chapter 9

Solution to Exercise 9.1

Fig. C.8 illustrates the decision framework of the consumer, which is as follows:

1. Before the beginning of the week (e.g., at 10pm of the previous week Sunday), the consumer decides (a) which contracts to sign spanning the

Table C.14 Solution to Exercise 8.9: forward contract purchases and selling prices

β	Contract 1 purchases (MW)	Contract 2 purchases (MW)	Selling price for client group 1 (\$/MWh)	Selling price for client group 2 (\$/MWh)
0	0	0	75	70
1	20	0	75	80
2	20	20	85	80
5	40	20	85	80
15	40	40	85	80

- whole week and (b) which contracts to sign spanning the working days of the week. These are here-and-now decisions.
- Before the beginning of Saturday (e.g., at 10 pm of Friday), the consumer decides which contracts to sign spanning the weekend. These are here-and-now decisions for weekend prices but wait-and-see decisions for prices pertaining to the working days.
 - Throughout the week the consumer trades in the pool. These are wait-and-see decisions.

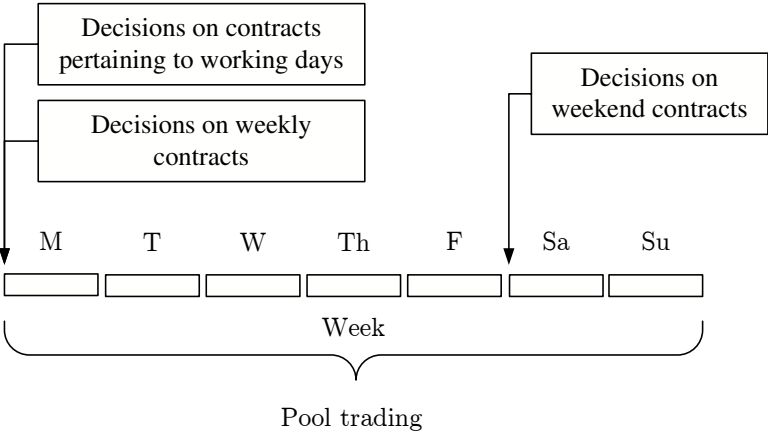


Fig. C.8 Solution to Exercise 9.1: decision framework for the consumer

Solution to Exercise 9.6

The two forward contracts described in Table C.15 are included as buying options for the consumer of the example in Section 9.5 of Chapter 9.

Table C.16 provides the expected cost and the CVaR for different values of the risk-aversion parameter β .

Table C.15 Solution to Exercise 9.6: forward contract data

Contract #	Usage period	Price	Upper bound
		(\$/MWh)	(MW)
1	1-4	48.5	30
2	1-4	50	30

Table C.16 Solution to Exercise 9.6: cost results

β	Expected cost (\$)	CVaR (\$)
0	44,073.44	46,525.00
1	44,328.59	45,622.50
2	44,938.75	45,197.50
10	44,938.75	45,197.50

Table C.17 provides the power purchased from bilateral and forward contracts. Observe that only those bilateral contracts decided at the beginning of the planning horizon and constituting therefore first-stage decision variables are reported (bilateral contracts 1, 2 and 3).

Table C.17 Solution to Exercise 9.6: power purchased from bilateral and forward contracts (MW)

β	Bilateral contracts			Forward contracts	
	1	2	3	1	2
0	0	0	0	0	0
1	75	0	0	30	0
2	75	65	0	30	30
10	75	65	0	30	30

By comparing, respectively, Tables C.16 and C.17 with Tables 9.4 and 9.5 in Chapter 9, the observations below are in order:

1. The incorporation of forward contracts reduces the risk faced by the consumer (CVaR) at the cost of increasing the expected cost.
2. The power purchased from forward contracts increases as the parameter β grows.
3. The power purchased from bilateral contracts 2 and 3 diminishes if forward contracts are considered.

Solution to Exercise 9.8

The considered reduced set of scenarios pertaining to the example in Section 9.5 of Chapter 9 are provided in Table C.18.

Table C.18 Solution to Exercise 9.8: pool price scenarios (\$/MWh)

Scenario #	Period #				Scenario #	Period #			
	1	2	3	4		1	2	3	4
1	44	44	42	42	5	50	52	50	51
2	44	44	42	45	6	50	52	50	55
3	44	44	44	45	7	50	52	51	52
4	44	44	44	46	8	50	52	51	53

Considering the scenarios in Table C.18, the results obtained are provided in Tables C.19 and C.20.

Table C.19 provides cost results, including expected cost, cost standard deviation and CVaR, while Table C.20 provides electricity procurement results, including expected contract purchases, expected pool purchases and sales, and expected production by the self-production unit.

Table C.19 Solution to Exercise 9.6: cost results

β	Expected cost (\$)	Cost standard deviation (\$)	CVaR (\$)
0	43,950.00	2302.21	46,500.00
1	44,085.94	1741.36	46,012.50
2	44,446.88	1182.06	45,750.00
10	44,509.38	1109.44	45,737.50

Table C.20 Solution to Exercise 9.6: expected electricity procurement

β	Expected contract purchases (MWh)	Expected pool purchases (MWh)	Expected pool sales (MWh)	Expected self-production (MWh)
0	75.00	662.50	0.00	212.50
1	375.00	362.50	0.00	212.50
2	675.00	162.50	100.00	212.50
10	725.00	137.50	125.00	212.50

By comparing, respectively, Tables C.19 and C.20 with Tables 9.4 and 9.6 in Chapter 9, the observations below are in order:

1. The results in terms of expected cost, cost standard deviation and CVaR for the case with 8 scenarios are very similar to those obtained with 16 scenarios.
2. In the case with 8 scenarios the purchases from bilateral contracts and the absolute value of the energy traded in the pool increase with respect to the original case.
3. The expected production of the self-production unit resulting from the case with 8 scenarios decreases with respect to the value obtained in the original case.

C.9 Exercises from Chapter 10

Solution to Exercises 10.1 and 10.2

- a) If sequential and compound failures are ignored, the number of failure scenarios N_Ω is given by

$$N_\Omega = N_T \times N_E + 1 = 4 \times 6 = 25,$$

where N_T is the number of periods spanning the scheduling horizon and N_E is the number of pieces of equipment (lines and generators) in the power system.

- b) If sequential failures are disregarded, but compound failures (defined as the simultaneous outage of several pieces of equipment) are taken into account, the number of failure scenarios is computed as

$$N_\Omega = N_T \times \sum_{i=1}^{N_E} \binom{N_E}{i} + 1 = N_T \times (2^{N_E} - 1) + 1 = 4 \times (2^6 - 1) + 1 = 253.$$

- c) If every piece of equipment in the system may fail at any time period of the scheduling horizon, the number of failure scenarios is

$$N_\Omega = \sum_{i=0}^{N_E} N_T^i \binom{N_E}{i} = \sum_{i=0}^6 4^i \binom{6}{i} = 15,625.$$

Solution to Exercise 10.3

According to the load shedding events provided in Table 10.15 of Chapter 10, the LOLP, LOLE, and ELNS associated with the scheduling program are

$$\text{LOLP} = \pi_1 + \pi_2 + \pi_6 + \pi_7 + \pi_8 = 2 \times 0.001 + 3 \times 0.005 = 0.017$$

$$\begin{aligned}\text{LOLE} &= 3 \text{ h } \pi_1 + 3 \text{ h } \pi_2 + 3 \text{ h } \pi_6 + 2 \text{ h } \pi_7 + 1 \text{ h } \pi_8 = 6 \text{ h} \times 0.001 + 6 \\ &\quad \text{h} \times 0.005 = 0.036 \text{ h} \\ \text{ELNS} &= (1 + 7 + 7) \text{ MWh } \pi_1 + (1 + 5 + 2) \text{ MWh } \pi_2 + (2+13+10) \text{ MWh} \\ &\quad \pi_6 + (5+4) \text{ MWh } \pi_7 + 2 \text{ MWh } \pi_8 = 23 \text{ MWh} \times 0.001 + 36 \text{ MWh} \times \\ &\quad 0.005 = 0.203 \text{ MWh}.\end{aligned}$$

Solution to Exercise 10.6

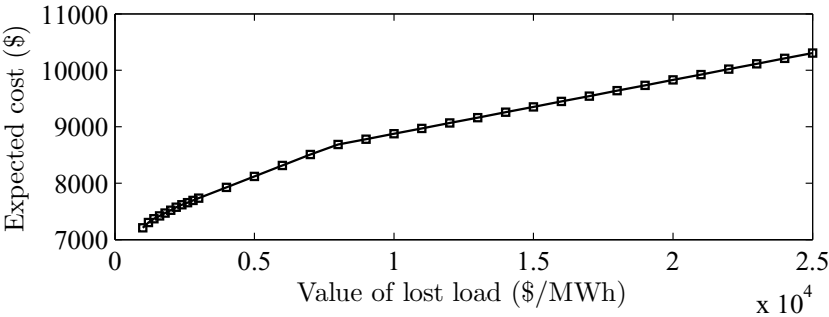
Fig. C.9 shows the evolution of the expected cost, the expected load not served (ELNS), and the total amount of reserves scheduled in the market as the value of lost load increases. Observe that the expected load not served remains different from zero for values of lost load as high as \$25,000/MWh. In particular, from a value of lost load higher than \$7825/MWh, the expected load not served remains constant and equal to its lowest possible value, which turns out to be 0.0954 MWh.

Table C.21 provides the power and reserve schedules resulting from the market-clearing procedure for a value of lost load equal to \$25,000/MWh. Note that the 11 MW of up-spinning reserve offered by the only load in the system in period 3 are accepted in the electricity market and consequently scheduled in that period. This way, the consumption level at node 3 can be voluntarily reduced to 99 MW in period 3. Even so, a part of the load still needs to be shed for technical reasons (see Table C.22 below) under every scenario ω such that the failure of unit 3 or the outage of line 2 takes place before period 4, i.e., under scenarios $\omega = 1, 2, 3, 5, 6$, and 7. If unit

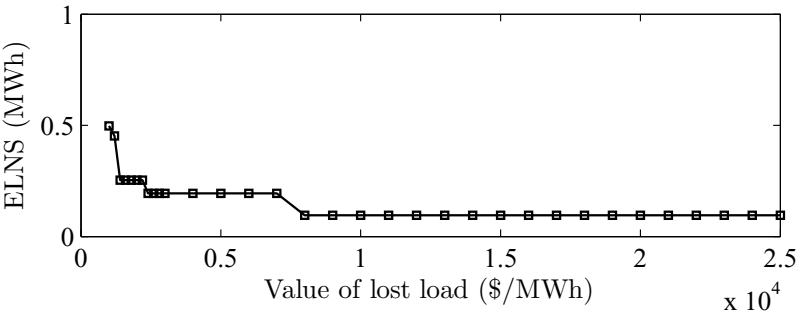
Table C.21 Solution to Exercise 10.6: scheduling program for $V^{\text{LOL}} = \$25,000/\text{MWh}$. Powers in MW

Period #	P_t^S			R_t^U				R_t^D				R_t^{NS}		
	G1	G2	G3	G1	G2	G3	L3	G1	G2	G3	L3	G1	G2	G3
1	0	0	30	0	0	0	0	0	0	0	0	30	0	0
2	30	0	50	50	0	0	0	0	0	0	0	0	0	0
3	60	0	50	15	0	0	11	25	0	0	0	0	15	0
4	0	0	40	0	0	0	0	0	0	0	0	40	0	0

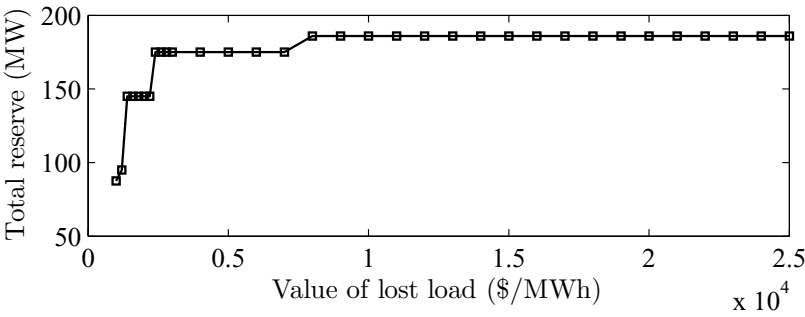
3 is forced out, the maximum load that can be supplied at node 3 is 90 MW. This power comes from generating units 1 and 2 through lines 2 and 3, whose transmission capacities are 55 and 35 MW, respectively. Therefore, if unit 3 becomes unavailable before period 4, 9 MW of demand need to be involuntarily curtailed in period 3. This situation is graphically represented in Fig. C.10. Likewise, if line 2 is forced out (but not unit 3), the maximum demand that can be satisfied at node 3 is 85 MW, 50 of which are produced by unit 3 located at that node. The remaining 35 MW come from unit 1 through lines 1 and 3. As a result, if the outage of line 2 occurs before period 4, 14 MW of load have to be necessarily shed in period 3. This situation is illustrated in Fig. C.11.



(a)



(b)



(c)

Fig. C.9 Solution to Exercise 10.6: impact of the value of lost load on: (a) the expected social cost; (b) the expected amount of load not served (ELNS); (c) the total amount of reserve scheduled in the market

Table C.22 Solution to Exercise 10.6: involuntary load shedding (MWh) for $V^{\text{LOL}} = \$25,000/\text{MWh}$

Period #	Scenario #								
	0	1	2	3	4	5	6	7	8
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	9	9	9	0	14	14	14	0
4	0	0	0	0	0	0	0	0	0

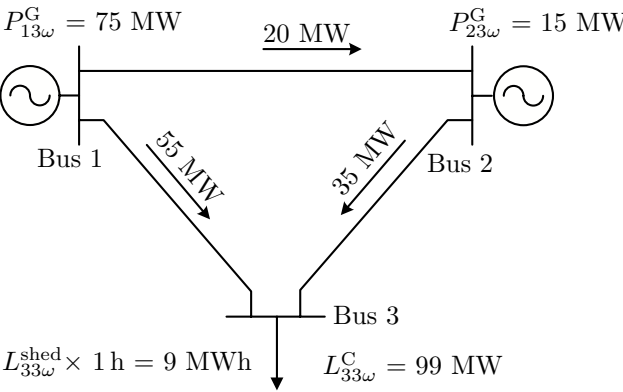


Fig. C.10 Solution to Exercise 10.6: effect of the outage of unit 3 during period $t = 3$ in scenarios $\omega = 1, 2$ and 3

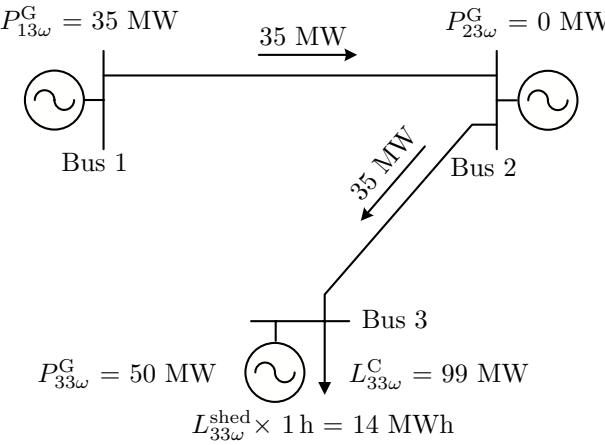


Fig. C.11 Solution to Exercise 10.6: effect of the outage of line 2 during period $t = 3$ in scenarios $\omega = 5, 6$ and 7

C.10 Exercises from Chapter 11

Solution to Exercise 11.2

Consider the following additional notation:

RD_i	Ramp-down limit of unit i .
RU_i	Ramp-up limit of unit i .
SD_i	Shut-down ramp limit of unit i .
SU_i	Start-up ramp limit of unit i .
D_i^T	Minimum down time of unit i .
U_i^T	Minimum up time of unit i .
S_i^0	Time periods unit i has been off-line at the beginning of the scheduling horizon.
V_i^0	Time periods unit i has been online at the beginning of the scheduling horizon.
u_i^0	Initial commitment status of unit i (1 online, 0 otherwise).

Following [20], the ramp-up and start-up ramp rates of each generating unit i are enforced by equations (C.58) below. Analogously, the shut-down ramp and ramp-down limits of each unit i are imposed through equations (C.59) and (C.60), respectively.

$$\begin{aligned}
 P_{it\omega}^G &\leq P_{i(t-1)\omega}^G + RU_i v_{i(t-1)\omega} \\
 &\quad + SU_i (v_{it\omega} - v_{i(t-1)\omega}) \\
 &\quad + P_i^{\max} (1 - v_{it\omega}), \quad \forall i, \forall t, \forall \omega
 \end{aligned} \tag{C.58}$$

$$\begin{aligned}
 P_{it\omega}^G &\leq P_i^{\max} v_{i(t+1)\omega} \\
 &\quad + SD_i (v_{it\omega} - v_{i(t+1)\omega}), \\
 &\quad \forall i, \forall t = 1, 2, \dots, N_T - 1, \forall \omega
 \end{aligned} \tag{C.59}$$

$$\begin{aligned}
 P_{i(t-1)\omega}^G - P_{it\omega}^G &\leq RD_i v_{it\omega} \\
 &\quad + SD_i (v_{i(t-1)\omega} - v_{it\omega}) \\
 &\quad + P_i^{\max} (1 - v_{i(t-1)\omega}), \quad \forall i, \forall \omega, \forall t.
 \end{aligned} \tag{C.60}$$

The minimum up-time constraints are written as follows:

$$\sum_{t=1}^{T_i^{\text{ON}}} (1 - v_{it\omega}) = 0, \quad \forall i, \quad \forall \omega \quad (\text{C.61})$$

$$\sum_{t_1=t}^{t+U_i^{\text{T}}-1} v_{it_1\omega} \geq U_i^{\text{T}}(v_{it\omega} - v_{i(t-1)\omega})$$

$$\forall i, \forall t = T_i^{\text{ON}} + 1, \dots, N_{\text{T}} - U_i^{\text{T}} + 1, \quad \forall \omega \quad (\text{C.62})$$

$$\sum_{t_1=t}^{N_{\text{T}}} [v_{it_1\omega} - (v_{it\omega} - v_{i(t-1)\omega})] \geq 0$$

$$\forall i, \forall t = N_{\text{T}} - U_i^{\text{T}} + 2, \dots, N_{\text{T}}, \quad \forall \omega, \quad (\text{C.63})$$

where T_i^{ON} is the number of initial periods during which unit i must be online. T_i^{ON} is mathematically expressed as

$$T_i^{\text{ON}} = \text{Min}\{N_{\text{T}}, (U_i^{\text{T}} - V_i^0) u_i^0\}.$$

Equations (C.61) constrain the initial status of the units according to T_i^{ON} . Equations (C.62) are enforced throughout the subsequent periods to comply with the minimum up time constraint during all the possible sets of consecutive periods of size U_i^{T} . Lastly, constraints (C.63) model the final $U_i^{\text{T}} - 1$ periods in which if unit i is started up, it remains online until the end of the time horizon.

Minimum down-time constraints are obtained from the minimum up-time constraints (C.61)–(C.63) if $v_{it\omega}$, U_i^{T} , and V_i^0 are replaced by $1 - v_{it\omega}$, D_i^{T} , and S_i^0 , respectively. That is,

$$\sum_{t=1}^{T_i^{\text{OFF}}} v_{it\omega} = 0, \quad \forall i, \quad \forall \omega \quad (\text{C.64})$$

$$\sum_{t_1=t}^{t+D_i^{\text{T}}-1} (1 - v_{it_1\omega}) \geq D_i^{\text{T}}(v_{i(t-1)\omega} - v_{it\omega})$$

$$\forall i, \forall t = T_i^{\text{OFF}} + 1, \dots, N_{\text{T}} - D_i^{\text{T}} + 1, \quad \forall \omega \quad (\text{C.65})$$

$$\sum_{t_1=t}^{N_{\text{T}}} [1 - v_{it_1\omega} - (v_{i(t-1)\omega} - v_{it\omega})] \geq 0$$

$$\forall i, \forall t = N_{\text{T}} - D_i^{\text{T}} + 2, \dots, N_{\text{T}}, \quad \forall \omega, \quad (\text{C.66})$$

where T_i^{OFF} is the number of initial periods during which unit i must be off-line. T_i^{OFF} is mathematically expressed as

$$T_i^{\text{OFF}} = \text{Min}\{N_{\text{T}}, (D_i^{\text{T}} - S_i^0) (1 - u_i^0)\}.$$

Fig. C.12 illustrates the evolution of the expected cost as the amount of wind generation increases. A 50% level of uncertainty, as defined in Section 11.5 of Chapter 11, is considered. Two instances are compared, namely, the base case and a new case study in which the unit ramping capabilities are reduced so that ramping constraints become binding. Observe that ramp limitations and network bottlenecks have a similar detrimental effect on the expected cost reduction achieved by wind integration.

In order to make ramping constraints binding, we can set, for example, the ramping limits of generating units to the values listed in Table. C.23.

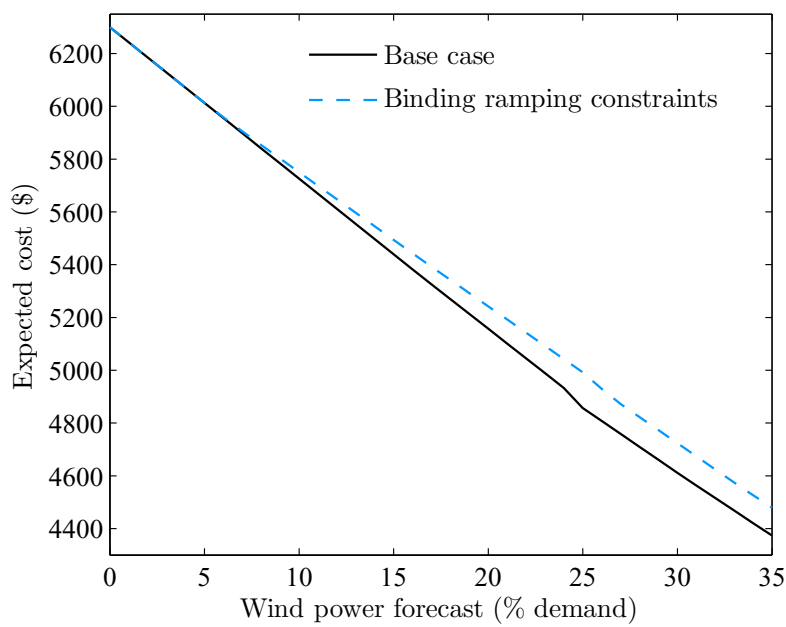


Fig. C.12 Solution to Exercise 11.2: effect of binding ramping constraints on the expected cost

Table C.23 Solution to Exercise 11.2: ramp limits in MW/h

Generator #	1	2	3
RD_i	20	90	12
RU_i	90	90	22
SD_i	100	100	50
SU_i	30	100	50

Solution to Exercise 11.6

Table C.24 provides the optimal production schedule and the optimal contracting of reserves with units and the load if decision variable $P_2^{\text{WP},S}$ is fixed to 20 MW. The results shown in square brackets for period 2 correspond to the base case described in Section 11.5 of Chapter 11, in which $P_2^{\text{WP},S} = 30$ MW. The results for the rest of periods are omitted because they are identical in both cases.

Table C.24 Solution to Exercise 11.6: scheduling program if $P_2^{\text{WP},S} = 20$ MW [30 MW]. Powers in MW

Period #	$P_t^{\text{WP},S}$	P_t^S			R_t^U				R_t^D				R_t^{NS}		
		G1	G2	G3	G1	G2	G3	L3	G1	G2	G3	L3	G1	G2	G3
1	6	0	0	24	0	0	4	0	0	0	0	0	0	0	0
	20	10	0	50	7	0	0	0	10	0	0	0	0	0	0
2	[30]	[0]	[0]	[50]	[0]	[0]	[0]	[0]	[0]	[0]	[0]	[0]	[17]	[0]	[0]
3	35	25	0	50	10	0	0	0	15	0	0	0	0	0	0
4	8	0	0	32	0	0	2	0	0	0	0	0	0	0	0

In order to satisfy the demand at bus 3 for any realization of the wind power production in period 2, the market operator contracts 17 MW of reserve with unit 1 for that period. If the market is cleared with a wind power schedule in period 2 of 20 MW, unit 1 is committed to take part in the energy dispatch in that period with 10 MW and consequently, the market operator has to contract those 17 MW of reserve from the spinning reserve offer of this unit. In contrast, if the market is cleared with a wind power schedule in period 2 of 30 MW, then unit 1 is out of the energy market. Consequently, the required 17 MW of reserve are obtained from the non-spinning reserve offer of this unit, which is cheaper. Thus, as can be observed in Table C.25, a lower expected cost is achieved by reducing the cost associated with the reserve scheduling.

Table C.25 Solution to Exercise 11.6: breakdown of expected social costs (\$)

$P_2^{\text{WP},S}$ (MW)	Total	Start-up	System operation	Reserve scheduling	Energy not served
20	4604.0	200	4146.0	258.0	0
30	4595.5	200	4146.0	249.5	0

Solution to Exercise 11.9

Table C.26 provides the scheduling program yielded by the market-clearing procedure described in Chapter 11 if the three wind power scenarios in Table 11.1 are assumed to be equiprobable and the load offers to sell both up- and down-spinning reserve at \$20/MWh. The results shown in square brackets for period 3 correspond to the network-congested case, in which the capacity of line 3 is reduced to 25 MW. The results of periods 1, 2 and 4 in the network-congested case are omitted because they are identical to the corresponding ones in the base case.

Table C.26 Solution to Exercise 11.9: scheduling program. Powers in MW

Period #	$P_t^{\text{WP}, S}$	P_t^S			R_t^U				R_t^D				R_t^{NS}		
		G1	G2	G3	G1	G2	G3	L3	G1	G2	G3	L3	G1	G2	G3
1	6	0	0	24	0	0	4	0	0	0	0	0	0	0	0
2	30	0	0	50	0	0	0	0	0	0	0	0	17	0	0
3	60	0	0	50	0	0	0	11	0	0	0	0	24	0	0
	[36]	[24]	[0]	[50]	[0]	[0]	[0]	[11]	[1]	[0]	[0]	[0]	[0]	[0]	[0]
4	8	0	0	32	0	0	2	0	0	0	0	0	0	0	0

Note that in the periods of highest demand (2 and 3), the hourly wind production scheduled in the market exceeds the wind power forecast in those periods, especially in the base case, for which the scheduled wind production in period 3 (60 MW) is even higher than the wind power value corresponding to the *high* scenario (50 MW). The reason for this is that clearing the market with a wind power value of 60 MW in period 3 allows the market operator to solely commit unit 3 (with a scheduled power of 50 MW) to meet the period load of 110 MW, provided that 24 MW of non-spinning reserve are contracted with unit 1 to guarantee the supply if the lowest values of wind power materialize. On the contrary, in the network-congested case (see results in square brackets) and for any realization of wind, unit 1 needs to be started up in order to supply the load at bus 3 by transferring power through line 2. Consequently, the market schedule includes a power generation of 24 MW for unit 1, while non-spinning reserves are not required.

References

1. T. Ackermann, J. Rivier Abbad, I. M. Dudurych, I. Erlich, H. Holttinen, J. Runge Kristoffersen, and Ejnar Sørensen. European Balancing Act. *IEEE power & energy magazine*, 5(6):91–103, 2007.
2. J. M. Angarita and J. G. Usaola. Combining hydro-generation and wind energy biddings and operation on electricity spot markets. *Electric Power Systems Research*, 77(5–6):393–400, 2007.
3. J. M. Arroyo and A. J. Conejo. Optimal response of a thermal unit to an electricity spot market. *IEEE Transactions on Power Systems*, 15(3):1098–1104, Aug. 2000.
4. J. M. Arroyo and A. J. Conejo. Multiperiod auction for a pool-based electricity market. *IEEE Transactions on Power Systems*, 17(4):1225–1231, 2002.
5. J. M. Arroyo and A. J. Conejo. Optimal response of a power generator to energy, AGC, and reserve pool-based markets. *IEEE Transactions on Power Systems*, 17(2):404–410, May 2002.
6. J. M. Arroyo and F. D. Galiana. Energy and reserve pricing in security and network-constrained electricity markets. *IEEE Transactions on Power Systems*, 20(2):634–643, 2005.
7. P. Artzner, F. Delbaen, J. M. Eber, and D. Heath. Coherent measures of risk. *Mathematical Finance*, 9(3):203–228, Jul. 1999.
8. A. Baflo, M. Ventosa, M. Rivier, and A. Ramos. Optimal offering strategies for generation companies operating in electricity spot markets. *IEEE Transactions on Power Systems*, 19(2):745–753, May 2004.
9. A. G. Bakirtzis, N. P. Ziogos, A. C. Tellidou, and G. A. Bakirtzis. Electricity producer offering strategies in day-ahead energy market with step-wise offers. *IEEE Transactions on Power Systems*, 22(4):1804–1818, Nov. 2007.
10. G. N. Bathurst and G. Strbac. Value of combining energy storage and wind in short-term energy and balancing markets. *Electric Power Systems Research*, 67(1):1–8, 2003.
11. G. N. Bathurst, J. Weatherill, and G. Strbac. Trading wind generation in short term energy markets. *IEEE Transactions on Power Systems*, 17(3):782–789, 2002.
12. R. Billinton and R. N. Allan. *Reliability Evaluation of Power Systems*. Plenum Press, New York and London, 1984.
13. R. Billinton and R. Karki. Application of Monte Carlo simulation to generating system well-being analysis. *IEEE Transactions on Power Systems*, 14(3):1172–1177, 1999.
14. J. R. Birge and F. Louveaux. *Introduction to stochastic programming*. Springer-Verlag, New York, 1997.

15. F. Bouffard and F. D. Galiana. Stochastic security for operations planning with significant wind power generation. *IEEE Transactions on Power Systems*, 23(2):306–316, 2008.
16. F. Bouffard, F. D. Galiana, and A. J. Conejo. Market-clearing with stochastic security—Part I: formulation. *IEEE Transactions on Power Systems*, 20(4):1818–1826, 2005.
17. F. Bouffard, F. D. Galiana, and A. J. Conejo. Market-clearing with stochastic security—Part II: case studies. *IEEE Transactions on Power Systems*, 20(4):1827–1835, 2005.
18. G. E. P. Box, G. M. Jenkins, and G. C. Reinsel. *Time Series Analysis: Forecasting and Control*. John Wiley & Sons, New Jersey, USA, 2008.
19. A. Brooke, D. Kendrick, and A. Meeraus. *Release 2.25 GAMS A User's Guide*. South San Francisco, 1992.
20. M. Carrión and J. M. Arroyo. A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem. *IEEE Transactions on Power Systems*, 21(3):1371–1378, 2006.
21. M. Carrión, A. B. Philpott, A. J. Conejo, and J. M. Arroyo. A stochastic programming approach to electric energy procurement for large consumers. *IEEE Transactions on Power Systems*, 22(2):744–754, May 2007.
22. E. D. Castronuovo and J. A. Peças Lopes. On the optimization of the daily operation of a wind-hydro power plant. *IEEE Transactions on Power Systems*, 19(3):1599–1606, 2004.
23. E. Çelebi and J. D. Fuller. A model for efficient consumer pricing schemes in electricity markets. *IEEE Transactions on Power Systems*, 22(1):60–67, Feb. 2007.
24. R. A. Collins. The economics of electricity hedging and a proposed modification for the futures contract for electricity. *IEEE Transactions on Power Systems*, 17(1):100–107, 2002.
25. A. J. Conejo and M. Carrión. Risk-constrained electricity procurement for a large consumer. *IEE Proceedings - Generation, Transmission and Distribution*, 153(4):407–413, Jul. 2006.
26. A. J. Conejo, E. Castillo, R. Mínguez, and R. García Bertrand. *Decomposition Techniques in Mathematical Programming. Engineering and Science Applications*. Springer, Heidelberg, Germany, 2006.
27. A. J. Conejo, J. Contreras, R. Espínola, and M. A. Plazas. Forecasting electricity prices for a day-ahead pool-based electric energy market. *International Journal of Forecasting*, 21(3):435–462, Jul.-Sep. 2005.
28. A. J. Conejo, R. García-Bertrand, M. Carrión, Á. Caballero, and A. de Andrés. Optimal involvement in forward markets of a power producer. *IEEE Transactions on Power Systems*, 23(2):703–711, 2008.
29. A. J. Conejo, F. J. Nogales, J. M. Arroyo, and R. García-Bertrand. Risk-constrained self-scheduling of a thermal power producer. *IEEE Transactions on Power Systems*, 19(3):1569–1574, Aug. 2004.
30. A. J. Conejo, F. J. Nogales, M. Carrión, and J. M. Morales. Electricity pool prices: Long-term uncertainty characterization for futures-market trading and risk management. *Journal of the Operational Research Society*, 61(2):235–245, Feb. 2010.
31. R. Dahlgren, C.-C. Liu, and J. Lawarrée. Risk assessment in energy trading. *IEEE Transactions on Power Systems*, 18(2):503–511, May 2003.
32. D. Das and B. F. Wollenberg. Risk assessment of generators bidding in day-ahead market. *IEEE Transactions on Power Systems*, 20(1):416–424, Feb. 2005.
33. S. de la Torre, J. M. Arroyo, A. J. Conejo, and J. Contreras. Price-maker self-scheduling in a pool-based electricity market: a mixed-integer LP approach. *IEEE Transactions on Power Systems*, 17(4):1037–1042, Nov. 2002.
34. E. A. DeMeo, W. Grant, M. R. Milligan, and M. J. Schuerger. Wind plant integration. *IEEE power & energy magazine*, 3(6):38–46, 2005.

35. E. A. DeMeo, G. A. Jordan, C. Kalich, J. King, M. R. Milligan, C. Murley, B. Oakleak, and M. J. Schuerger. Accommodating wind's natural behavior. *IEEE power & energy magazine*, 5(6):68–77, 2007.
36. S. J. Deng and S. S. Oren. Electricity derivatives and risk management. *Energy*, 31(6-7):940953, May-June 2006.
37. E. Denny and M. O'Malley. Wind generation, power system operation, and emissions reduction. *IEEE Transactions on Power Systems*, 21(1):341–347, 2006.
38. E. Denny and M. O'Malley. Quantifying the total net benefits of grid integrated wind. *IEEE Transactions on Power Systems*, 20(2):605–615, 2007.
39. D. Dentcheva and A. Ruszczyński. Portfolio optimization with stochastic dominance constraints. *Journal of Banking & Finance*, 30(2):433–451, 2006.
40. D. Dentcheva and A. Ruszczyński. Optimization with stochastic dominance constraints. *SIAM Journal on Optimization*, 14(2):548–566, Feb. 2003.
41. T. S. Dillon, K. W. Edwin, and R. J. Taud. Integer programming approach to the problem of optimal unit commitment with probabilistic reserve determination. *IEEE Transactions on Power Apparatus and Systems*, PAS-97(6):2154–2166, 1978.
42. R. Doherty and M. O'Malley. A new approach to quantify reserve demand in systems with significant installed wind capacity. *IEEE Transactions on Power Systems*, 20(2):587–595, 2005.
43. J. Dupačová, G. Consigli, and S. W. Wallace. Scenarios for multistage stochastic programs. *Annals of Operations Research*, 100(1-4):25–53, Dec. 2000.
44. J. Dupačová, N. Gröwe-Kuska, and W. Römisch. Scenario reduction in stochastic programming: An approach using probability metrics. *Mathematical Programming Series A*, 95(3):493–511, Mar. 2003.
45. Electrical and Computer Engineering of Kansas State University. Dr. Gary Johnson's renewable energy and tesla coil page.
Available at <http://eece.ksu.edu/~gjohnson>, 2010.
46. L. F. Escudero, A. Garín, M. Merino, and G. Pérez. The value of the stochastic solution in multistage problems. *TOP*, 15(1):48–64, July 2007.
47. European Energy Exchange, EEX.
Available at <http://www.eex.com>, 2010.
48. European Union. Directive 96/92/EC.
Available at <http://ec.europa.eu/energy>, 2010.
49. A. Gómez Exposito, A. J. Conejo, and C. Cañizares. *Electric Energy Systems: Analysis and Operation*. CRC Press, Taylor and Francis, Boca Raton, Florida, 2008.
50. FERC, Federal Energy Regulatory Commission, US.
Available at <http://www.ferc.gov>, 2010.
51. S.-E. Fleten and T. K. Kristoffersen. Stochastic programming for optimizing bidding strategies of a Nordic hydropower producer. *European Journal of Operational Research*, 181(2):916–928, Sept. 2007.
52. S.-E. Fleten and J. Lemming. Constructing forward price curves in electricity markets. *Energy Economics*, 25(5):409–424, Sep. 2003.
53. M. Flynn, W. P. Sheridan, J. D. Dillon, and M. J. O'Malley. Reliability and reserve in competitive electricity market scheduling. *IEEE Transactions on Power Systems*, 16(1):78–87, 2001.
54. Reliability Test System Task Force. The IEEE Reliability Test System—1996. *IEEE Transactions on Power Systems*, 14(3):1010–1020, 1999.
55. S. A. Gabriel, A. J. Conejo, M. A. Plazas, and S. Balakrishnan. Optimal price and quantity determination for retail electric power contracts. *IEEE Transactions on Power Systems*, 21(1):180–187, Feb. 2006.
56. S. A. Gabriel, M. F. Genc, and S. Balakrishnan. A simulation approach to balancing annual risk and reward in retail electrical power markets. *IEEE Transactions on Power Systems*, 17(4):1050–1057, Nov. 2002.
57. F. D. Galiana and F. Bouffard. An electricity market with a probabilistic spinning reserve criterion. *IEEE Transactions on Power Systems*, 19(1):300–307, 2004.

58. F. D. Galiana, F. Bouffard, J. M. Arroyo, and J. F. Restrepo. Scheduling and pricing of coupled energy and primary, secondary, and tertiary reserves. *Proceedings of the IEEE*, 93(11):1970–1983, 2005.
59. L. Garcés and A. J. Conejo. Weekly self-scheduling, forward contracting and offering strategy for a producer. *IEEE Transactions on Power Systems*, 25(2):657–666, May 2010.
60. A. Garcia, J. L. Torres, E. Prieto, and A. de Francisco. Fitting wind speed distributions: a case study. *Solar Energy*, 62(2):139–144, 1998.
61. J. García-González, R. Moraga Ruiz de la Muela, L. Matres Santos, and A. Mateo González. Stochastic joint optimization of wind generation and pumped-storage units in an electricity market. *IEEE Transactions on Power Systems*, 23(2):460–468, 2008.
62. T. W. Gedra. Optional forward contracts for electric power markets. *IEEE Transactions on Power Systems*, 9(4):1766–1773, 1994.
63. H. A. Gil and G. Joos. Generalized estimation of average displaced emissions by wind generation. *IEEE Transactions on Power Systems*, 22(3):1035–1043, 2007.
64. Global Wind Energy Council. Global wind 2009 report. Available at <http://www.gwec.net>, 2010.
65. R. Gollmer, U. Gotzes, and R. Schultz. Second-order stochastic dominance constraints induced by mixed-integer linear recourse. *Preprint 644-2007, Department of Mathematics, University of Duisburg-Essen*, 2007.
66. R. Gollmer, F. Neise, and R. Schultz. Stochastic programs with first-order dominance constraints induced by mixed-integer linear recourse. *Preprint 641-2006, Department of Mathematics, University of Duisburg-Essen*, 2006.
67. E. Gómez-Villalva and A. Ramos. Optimal energy management of an industrial consumer in liberalized markets. *IEEE Transactions on Power Systems*, 18(2):716–723, May 2003.
68. H. B. Gooi, D. P. Mendes, K. R. W. Bell, and D. S. Kirschen. Optimal scheduling of spinning reserve. *IEEE Transactions on Power Systems*, 14(4):1485–1492, 1999.
69. V. P. Gountis and A. G. Bakirtzis. Bidding strategies for electricity producers in a competitive electricity marketplace. *IEEE Transactions on Power Systems*, 19(1):356–365, Feb. 2004.
70. J. D. Guy. Security constrained unit commitment. *IEEE Transactions on Power Apparatus and Systems*, PAS-90(3):1385–1390, 1971.
71. R. P. Härmäläinen, J. Mäntysaari, J. Ruusunen, and P.-O. Pineau. Cooperative consumers in a deregulated electricity market – dynamic consumption strategies and price coordination. *Energy*, 25(9):857–875, Sep. 2000.
72. A. C Harvey. *Time Series Models*. John Wiley & Sons Inc, New York, USA, 1981.
73. K. Hedman and G. Sheble. Comparing hedging methods for wind power: using pumped storage hydro units vs options purchasing. *International Conference on Probabilistic Methods Applied to Power Systems, PMAPS'06, Stockholm, Sweden*.
74. H. Heitsch and W. Römis. Scenario reduction algorithms in stochastic programming. *Computational Optimization and Applications*, 24(2-3):187–206, Feb.-Mar. 2003.
75. H. Heitsch and W. Römis. A note on scenario reduction for two-stage stochastic programs. *Operations Research Letters*, 35:731–738, 2007.
76. J. P. Hennessey. Some aspects of wind power statistics. *Journal of Applied Meteorology*, 16(2):119–128, 1977.
77. S. W. Heunis and R. Herman. A probabilistic model for residential consumer loads. *IEEE Transactions on Power Systems*, 17(3):621–625, Aug. 2002.
78. J. L. Hige. *Tutorials in Operations Research, INFORMS 2005. Chapter 2: Stochastic Programming: Optimization When Uncertainty Matters*. INFORMS, Hanover, Maryland, 2005.

79. J. L. Higle and S. Sen. Stochastic decomposition – an algorithm for 2-stage linear-programs with recourse. *Mathematics of Operations Research*, 16(3):650–669, Aug. 1991.
80. J. L. Higle and S. Sen. *Stochastic Decomposition. A Statistical Method for Large Scale Stochastic Linear Programming*. Springer, New York, New York, 1996.
81. H. Hotelling. New light on the correlation coefficient and its transforms. *Journal of the Royal Statistical Society Series B-Statistical Methodology*, 15(2):193–232, 1953.
82. K. Høyland, M. Kaut, and S. W. Wallace. A heuristic for moment-matching scenario generation. *Computational Optimization and Applications*, 24(2-3):169–185, Feb. 2003.
83. K. Høyland and S. W. Wallace. Generating scenario trees for multistage decision problems. *Management Science*, 47(2):295–307, Feb. 2001.
84. P. Kall and S. W. Wallace. *Stochastic Programming*. John Wiley and Sons, Chichester, UK, 1994.
85. K. K. Kariuki and R. N. Allan. Evaluation of reliability worth and value of lost load. *IEEE Proceedings-Generation, Transmission and Distribution*, 143(2):171–180, 1996.
86. M. Kaut and S. W. Wallace. Evaluation of scenario-generation methods for stochastic programming. *Pacific Journal of Optimization*, 3(2):257–271, 2007.
87. R. J. Kaye, H. R. Outhred, and C. H. Bannister. Forward contracts for the operation of an electricity industry under spot pricing. *IEEE Transactions on Power Systems*, 5(2):46–52, 1990.
88. D. S. Kirschen and G. Strbac. *Fundamentals of Power System Economics*. John Wiley & Sons, Ltd., Chichester, West Sussex, UK, 2004.
89. P.-L. Liu and A. Der Kiureghian. Multivariate distribution models with prescribed marginals and covariances. *Probabilistic Engineering Mechanics*, 1(2):105–112, 1986.
90. Y. F. Liu and F. F. Wu. Generator bidding in oligopolistic electricity markets using optimal control: Fundamentals and application. *IEEE Transactions on Power Systems*, 21(3):1050–1061, Aug. 2006.
91. H. M. Markowitz. *Portfolio Selection: Efficient Diversification of Investments*. Wiley, New York, 1959.
92. A. Mas-Colell, M. D. Whinston, and J. R. Green. *Microeconomic Theory*. Oxford University Press, New York, 1995.
93. J. Matevosyan and L. Söder. Minimization of imbalance cost trading wind power on the short-term power market. *IEEE Transactions on Power Systems*, 21(3):1396–1404, 2006.
94. J. M. Morales, A. J. Conejo, and J. Pérez-Ruiz. Economic valuation of reserves in power systems with high penetration of wind power. *IEEE Transactions on Power Systems*, 24(2):900–910, 2009.
95. J. M. Morales, A. J. Conejo, and J. Pérez-Ruiz. Short-term trading for a wind power producer. *IEEE Transactions on Power Systems*, 25(1):554–564, 2010.
96. J. M. Morales, R. Mínguez, and A. J. Conejo. A methodology to generate statistically dependent wind speed scenarios. *Applied Energy*, 87(3):843–855, Mar. 2010.
97. J. M. Morales, S. Pineda, A. J. Conejo, and M. Carrión. Scenario reduction for futures market trading in electricity markets. *IEEE Transactions on Power Systems*, 24(2):878–888, 2009.
98. A. Müller and D. Stoyan. *Comparison methods for stochastic models and risks*. Wiley, Chichester, 2002.
99. G. L. Nemhauser and L. A. Wolsey. *Integer and Combinatorial Optimization*. Wiley-Interscience, New York, 1999.
100. New York Electricity Market, NY ISO.
Available at <http://www.nyiso.com>, 2010.
101. New York Mercantile Exchange, NYMEX.
Available at <http://www.nymex.com>, 2010.

102. E. Ni, P. B. Luh, and S. Rourke. Optimal integrated generation bidding and scheduling with risk management under a deregulated power market. *IEEE Transactions on Power Systems*, 19(1):600–609, Feb. 2004.
103. W. Nicholson. *Microeconomic Theory. Basic Principles and Extensions*. Dryden Press, New York, ninth edition, 2005.
104. H. Niu, R. Baldick, and G. D. Zhu. Supply function equilibrium bidding strategies with fixed forward contracts. *IEEE Transactions on Power Systems*, 20(4):1859–1867, 2005.
105. W. Ogryczak and A. Ruszczyński. From stochastic dominance to mean-risk models: Semideviations as risk measures. *European Journal of Operational Research*, 116(1):33–50, 1999.
106. OMEL, Market Operator of the Electricity Market of Mainland Spain. Available at <http://www.omel.es>, 2010.
107. OMP, Futures Market of the Iberian Electricity Market. available at <http://www.omip.pt>, 2010.
108. IEEE/CIGRE Joint Task Force on Stability Terms and Definitions. Definition and clasification of power system stabiliy. *IEEE Transactions on Power Systems*, 19(2):1387–1401, 2004.
109. M. A. Ortega-Vazquez and D. S. Kirschen. Estimating the spinning reserve requirements in systems with significant wind power generation penetration. *IEEE Transactions on Power Systems*, 24(1):114–124, 2009.
110. D. Peña, G. C. Tiao, and R. S. Tsay (Eds). *A Course in Time Series Analysis*. John Wiley, New York, 2001.
111. Pennsylvania–New Jersey–Maryland Electricity Market, PJM ISO. Available at <http://www.pjm.com>, 2010.
112. M. V. Pereira, S. Granville, L. A. Barroso, M. Fampa, and R. Dix. Strategic bidding under uncertainty: A binary expansion approach. *IEEE Transactions on Power Systems*, 20(1):180–188, Feb. 2005.
113. E. Pettersen, A. B. Philpott, and S. W. Wallace. An electricity market game between consumers, retailers and network operators. *Decision Support Systems*, 40(3-4):427–438, Oct. 2005.
114. G. C. Pflug. Scenario tree generation for multiperiod financial optimization by optimal discretization. *Mathematical Programming Series B*, 89(2):251–271, Jan. 2001.
115. A. B. Philpott and E. Pettersen. Optimizing demand-side bids in day-ahead electricity markets. *IEEE Transactions on Power Systems*, 21(2):488–498, May 2006.
116. S. Pineda, A. J. Conejo, and M. Carrión. Impact of unit failure on forward contracting. *IEEE Transactions on Power Systems*, 23(4):1768–1775, 2008.
117. P. Pinson, C. Chevallier, and G. N. Kariniotakis. Trading wind generation from short-term probabilistic forecasts of wind power. *IEEE Transactions on Power Systems*, 22(3):1148–1156, 2007.
118. M. A. Plazas, A. J. Conejo, and F. J. Prieto. Multimarket optimal bidding for a power producer. *IEEE Transactions on Power Systems*, 20(4):2041–2050, Nov. 2005.
119. A. Prékopa. *Stochastic Programming*. Springer, New York, New York, 1995.
120. S. T. Rachev. *Probability Metrics and the Stability of Stochastic Models*. John Wiley & Sons, Chichester, England, 1991.
121. N. S. Rau. Optimal dispatch of a system based on offers and bids—a mixed integer lp formulation. *IEEE Transactions on Power Systems*, 14(1):274–279, 1999.
122. Red Eléctrica de España, S.A. Sistema de Información del Operador del Sistema, e-sios. Available at <http://www.esios.ree.es>, 2010.
123. RERL Wind Data. Renewable Energy Research Laboratory. Center for Energy Efficiency & Renewable Energy. University of Massachusetts at Amherst. Available at http://www.ceere.org/rerl/publications/resource_data/index.html, 2010.

124. R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.
125. R. T. Rockafellar and S. Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, Jul. 2002.
126. C. Ruiz and A. J. Conejo. Pool strategy of a producer with endogenous formation of locational marginal prices. *IEEE Transactions on Power Systems*, 24(4):1855–1866, Nov. 2009.
127. A. Ruszczyński and A. Shapiro (eds.). *Handbooks of Operations Research and Management Science. Stochastic Programming, vol. 10*. Elsevier, Amsterdam, Netherlands, 2003.
128. M. J. Schervish. P values: What they are and what they are not. *American Statistician*, 50(3):203–206, 1996.
129. R. Schultz and S. Tiedemann. Risk aversion via excess probabilities in stochastic programs with mixed-integer recourse. *Mathematical Finance*, 14(1):115–138, 2003.
130. F. C. Schweppe, M. C. Caramanis, R. D. Tabors, and R. E. Bohn. *Spot Pricing of Electricity*. Springer, 1988.
131. S. Sen and J. L. Higle. An introductory tutorial on stochastic linear programming: I (modeling). *Interfaces*, 29(2):33–61, March–April 1999.
132. M. Shahidehpour, T. Li, and Z. Li. Risk-constrained bidding strategy with stochastic unit commitment. *IEEE Transactions on Power Systems*, 22(1):449–458, Feb. 2007.
133. M. Shahidehpour, H. Yamin, and Z. Li. *Market Operations in Electric Power Systems: Forecasting, Scheduling, and Risk Management*. John Wiley and Sons, New York, New York, 2002.
134. M. Shahidehpour, H. Yamin, and Z. Li. *Market Operations in Electric Power Systems: Forecasting, Scheduling, and Risk Management*. IEEE-Wiley, New York, 2002.
135. G. B. Sheblé. *Computational Auction Mechanisms for Restructured Power Industry Operation*. Springer, New York, New York, 1999.
136. G. B. Shrestha, B. K. Pokharel, T. T. Lie, and S.-E. Fleten. Medium term power planning with bilateral contracts. *IEEE Transactions on Power Systems*, 20(2):627–633, 2005.
137. H. Singh and A. Papalexopoulos. Competitive procurement of ancillary services by an independent system operator. *IEEE Transactions on Power Systems*, 14(2):498–504, 1999.
138. S. Stoft. *Power System Economics: Designing Markets for Electricity*. John Wiley & Sons – IEEE Press, New York, New York, 2002.
139. G. Strbac, S. Ahmed, D. S. Kirschen, and R. N. Allan. A method for computing the value of corrective security. *IEEE Transactions on Power Systems*, 13(3):1096–1102, Aug. 1998.
140. E. Tanlapco, J. Lawarree, and C. C. Liu. Hedging with futures contracts in a deregulated electricity industry. *IEEE Transactions on Power Systems*, 17(3):577–582, 2002.
141. The GAMS Development Corporation Website.
Available at <http://www.gams.com>, 2010.
142. The ILOG CPLEX Website.
Available at <http://www.ilog.com/products/cplex/>, 2010.
143. K. Zare, A. J. Conejo, M. Carrión, and M. P. Moghaddam. Multi-market energy procurement for a large consumer using a risk-aversion procedure. *Electric Power Systems Research*, 80(1):63–70, Jan. 2010.

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